Journal Pre-proof



A Calibrated Spatio-Temporal Machine Learning Model for Tropical Peatland Fire Risk

Adilan W. Mahdiyasa, Jeremy F. Chaim, Udjianna S. Pasaribu, Elham Sumarga, Bagus P. Muljadi, Asrini Chrysanti

PII: S2666-5921(26)00072-7

DOI: https://doi.org/10.1016/j.nhres.2026.07.007

Reference: NHRES 352

To appear in: *Natural Hazards Research*

Received Date: 15 June 2026

Revised Date: 27 July 2026

Accepted Date: 28 July 2026

Please cite this article as: Mahdiyasa, A.W., Chaim, J.F., Pasaribu, U.S., Sumarga, E., Muljadi, B.P., Chrysanti, A., A Calibrated Spatio-Temporal Machine Learning Model for Tropical Peatland Fire Risk, *Natural Hazards Research*, https://doi.org/10.1016/j.nhres.2026.07.007.

This is a PDF of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability. This version will undergo additional copyediting, typesetting and review before it is published in its final form. As such, this version is no longer the Accepted Manuscript, but it is not yet the definitive Version of Record; we are providing this early version to give early visibility of the article. Please note that Elsevier's sharing policy for the Published Journal Article applies to this version, see: https://www.elsevier.com/about/policies-and-standards/sharing#4-published-journal-article. Please also note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2026 Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd.

Journal Pre-proof

A Calibrated Spatio-Temporal Machine Learning Model for Tropical Peatland Fire Risk

Adilan W. Mahdiyasa[1]*, Jeremy F. Chaim[2], Udjianna S. Pasaribu[1], Elham Sumarga[3], Bagus P. Muljadi[4] , Asrini Chrysanti[5]

[1]Department of Mathematics, Faculty of Mathematics and Natural Sciences, Bandung Institute of Technology, Bandung 40132, Indonesia

[2]Department of Actuarial Science, Faculty of Mathematics and Natural Sciences, Bandung Institute of Technology, Bandung 40132, Indonesia

[3]School of Life Sciences and Technology, Bandung Institute of Technology, Bandung 40132, Indonesia

[4]Department of Chemical and Environmental Engineering, Faculty of Engineering, University of Nottingham, University Park, Nottingham NG7 2RD, UK

[5]Department of Civil Engineering, Faculty of Civil and Environmental Engineering, Bandung Institute of Technology, Bandung 40132, Indonesia

* Corresponding author

adilan.widyawan@itb.ac.id (Adilan W. Mahdiyasa)

Journal Pre-proof

Journal Pre-proof

Abstract

Tropical peatland fires are complex environmental hazards with significant impacts. However, existing fire risk approaches, including empirical drought-fire indices and machine learning models that focus on either spatial or temporal aspects, remain limited in representing the dynamics of fire occurrence across heterogeneous peatland landscapes. Here, we develop an integrated spatio-temporal machine learning model with a probabilistic calibration method to analyze and predict tropical peatland fire risk. The framework employs a Convolutional Neural Network (CNN) to extract spatial features, Convolutional Long Short-Term Memory (ConvLSTM) to capture temporal dependencies, and isotonic regression to calibrate predicted fire probabilities. We employ meteorological, hydrological, land cover, anthropogenic, and historical fire data from tropical peatlands in Kalimantan, Indonesia. The proposed model achieves good performance and produces probability of fire maps that are consistent with observed fire hotspot distributions. Furthermore, the forecasting results show that the model can identify shifting fire risk patterns across peatland areas over different time horizons. These findings demonstrate that the proposed framework provides not only a predictive tool but also actionable information for early warning, risk assessment, and fire mitigation in tropical peatland ecosystems.

Keywords: Tropical peatland, Fire risk, Spatio-temporal model, Probability of fire, Machine learning model

Journal Pre-proof

1. Introduction

Tropical peatland fires are significant environmental hazards influenced by complex processes that evolve simultaneously across space and time. Analyzing and forecasting tropical peatland fire risk, including the location and timing of fire occurrence, can support risk assessment and fire mitigation efforts. This paper develops an integrated spatio-temporal machine learning model with a probabilistic calibration method to analyze and predict tropical peatland fire risk. The proposed framework combines three complementary methods. Convolutional Neural Network (CNN) is used to learn spatial patterns and identify areas with higher fire susceptibility. Convolutional Long Short-Term Memory (ConvLSTM) is applied to capture temporal dependencies and represent the evolution of fire prone conditions over time. Finally, isotonic regression is employed to calibrate the predicted fire probabilities, improving the consistency between model outputs and observed fire occurrence frequencies. Through this approach, the proposed model provides calibrated probability maps for examining and forecasting tropical peatland fire risk that are in agreement with observed fire hotspot distributions.

Tropical peatlands are critical ecosystems that contribute significantly to carbon storage, hydrological regulation, and the support of biodiversity (Alexandrov et al., 2020; Ledger et al., 2024; Loisel et al., 2021). Saturated conditions in tropical peatlands, associated with low hydraulic gradients (Baird et al., 2017), promote high rates of organic matter accumulation, resulting in the development of carbon rich ecosystems and contributing significantly to the global carbon cycle (Page et al., 2022). The high carbon storage capacity of undisturbed peat ecosystems limits greenhouse gas emissions, thereby contributing to climate change mitigation. By maintaining hydrological balance, tropical peatlands regulate water dynamics and reduce the intensity of hydrological extremes. The high water retention of tropical peatlands allows excess rainfall to be retained during wet periods and gradually

68 released during dry sea sons, which mitigates floods and maintains water availability (Lupascu
69 et al., 2020; Mukhaiyar et al., 2024; Wu et al., 2026). Furthermore, the unique environmental
70 conditions of tropical peatlands support habitats for biodiversity of high conservation value
71 (Girkin et al., 2022). A primary example is the endangered orangutan, Asia's only remaining
72 great ape, whose survival heavily depends among others on peat swamp forests (Felton et al.,
73 2003).

74 Although tropical peatlands provide important functions to the environment, they are
75 susceptible to severe and persistent fires that result in extensive impacts across multiple
76 interconnected sectors. Recurrent fires degrade peat structure, reduce biodiversity, and disrupt
77 hydrological processes that are critical to ecosystem resilience (Horton et al., 2022;
78 Mezbahuddin et al., 2023; Wang et al., 2022). Large scale fire events release massive amounts
79 of greenhouse gases and fine particulate matter into the atmosphere, contributing to climate
80 change and severe air pollution (Rodríguez Vásquez et al., 2021). The resulting haze is
81 associated with an increased incidence of respiratory and cardiovascular diseases across
82 affected regions (Koplitz et al., 2016; Kuwata, 2024), and contributes to both infant and adult
83 mortality (Hein et al., 2022). Moreover, tropical peatland fires produce significant financial
84 losses due to agricultural damage, infrastructure destruction, and supply chain interruption
85 (Kiely et al., 2021; Situngkir et al., 2025). The estimated economic loss in Indonesia resulting
86 from the 2015 tropical peatland fires was approximately US$16 billion, equivalent to 1.8% of
87 the country's GDP (Horton et al., 2021).

88 To support fire prevention strategies, drought-fire index models for tropical peatland
89 have been developed (Mashuri et al., 2026; Taufik et al., 2022, 2015). For example, Taufik et
90 al. (2022) proposed Peat Fire Vulnerability Index (PFVI), which incorporates hydrological and
91 meteorological variables, including water table, soil moisture, rainfall, and air temperature. The
92 PFVI provides interpretable metrics for assessing fire risk and classifies fire danger into four

93 categories. However, it remains limited by threshold-based structures, empirical parameters,
94 and partial representation of nonlinear interactions among variables. Consequently, the PFVI
95 may not effectively represent the complex feedback that govern fire occurrence across
96 heterogeneous tropical peatland landscapes.

97 To address these limitations, machine learning approaches are employed to model
98 complex interactions and improve the predictive accuracy of fire occurrence. Machine learning
99 methods provides flexibility in capturing nonlinear relationships and interactions among
100 multiple drivers without relying on predefined functional forms (Fauziah et al., 2025; Guria et
101 al., 2024, 2025; Horton et al., 2021; Kurniawan et al., 2025; Mahdiyasa et al., 2025;
102 Mezbahuddin et al., 2023; M. Mishra et al., 2024, 2025; Nie et al., 2026). However, the existing
103 machine learning models of tropical peatland fire risk focus on either temporal or spatial
104 aspects separately. For example, Mahdiyasa et al. (2025) developed a model *peatfr* to predict
105 and analyze tropical peatland fire through Long Short-Term Memory and Gate Recurrent Unit
106 machine learning methods. This model focuses primarily on temporal patterns and captures
107 time dependent relationships associated with fire occurrence. However, *peatfr* provides limited
108 information about the spatial distribution of fire risk. In contrast, Horton et al. (2021) proposed
109 a fire risk model for tropical peatland using machine learning methods XGBoost and Random
110 Forest. Although this approach successfully captured the key factors contributing to fire
111 susceptibility, the model focuses mainly on spatial characteristics and does not explicitly
112 account for temporal dynamics. Furthermore, the probability map produced by this model is
113 not properly calibrated and may overestimate the likelihood of fire occurrence, which reduces
114 the reliability of the model.

115 In this paper, we develop a novel spatio-temporal machine learning model with a
116 probabilistic calibration method to analyze and forecast tropical peatland fire risk. The
117 proposed model integrates Convolutional Neural Networks (CNN) to extract spatial features,

Convolutional Long Short-Term Memory (ConvLSTM) to learn temporal dependencies, and isotonic regression to calibrate the predicted probabilities of fire occurrence. Meteorological, hydrological, land cover, and anthropogenic variables are incorporated to capture the complex interactions that influence fire occurrence. This integrated approach aims to represent the processes governing fire ignition and propagation properly, which in turn support effective early warning systems and fire prevention strategies.

2. Methods

2.1 Study area

The study area is located in Kalimantan, Indonesia, which contains one of the largest tropical peatland ecosystems in Southeast Asia (**Figure 1**). Peatlands in this region cover approximately 4.54 million hectares, representing about 33.8% of total Indonesian peatlands (Anda et al., 2021). These peatlands are distributed in all five provinces, with the largest extents found in Central and West Kalimantan provinces. Peat depth in Kalimantan varies considerably, ranging from shallow deposits (around 50 cm) to deep peat layers (more than 700 cm) (Anda et al., 2021). Shallow peat deposits are generally found near peatland margins or transitional zones, while deeper peat layers are associated with central peat dome formations where long-term organic accumulation has occurred. The peatland distribution data employed in this study were obtained from the Geospatial Information Agency of Indonesia (BIG) (Geospatial Information Agency, 2019).

Journal Pre-proof

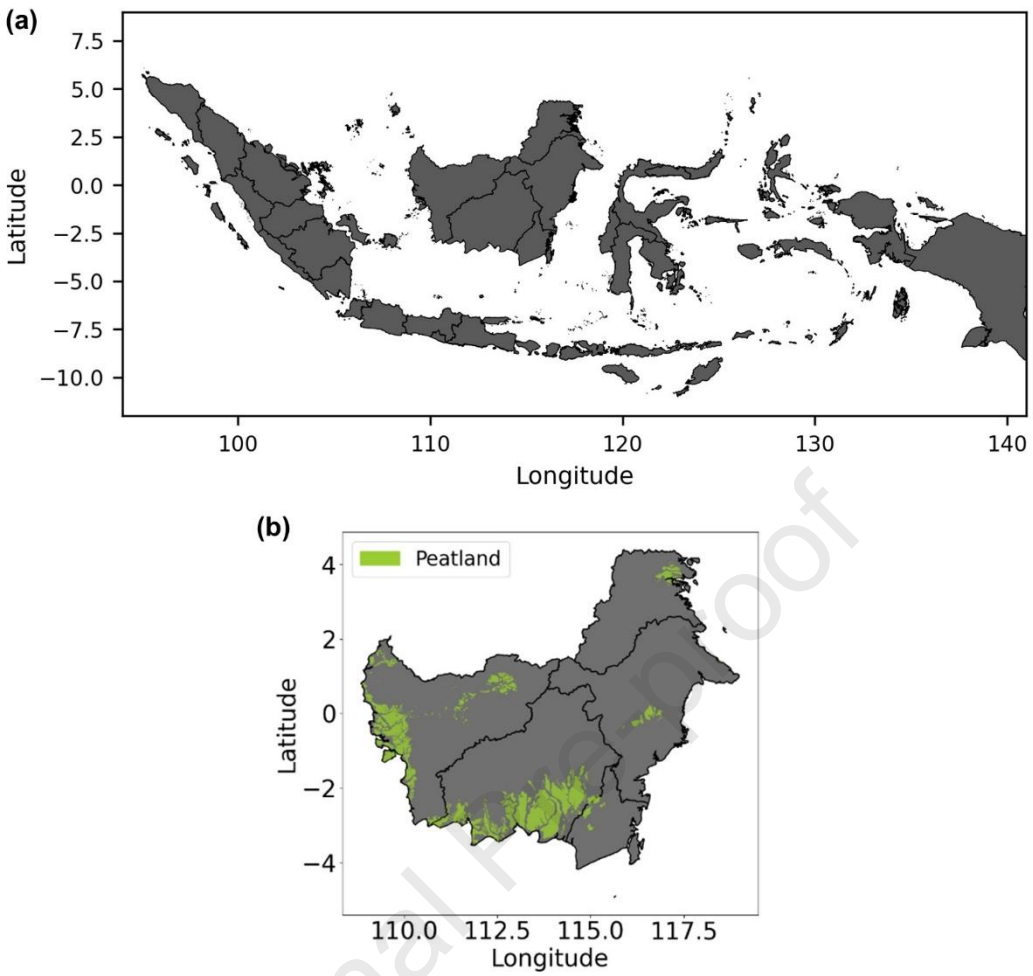


Figure 1. Location of the study area. (a) Map of Indonesia showing the geographic position of Kalimantan Island and (b) the distribution of tropical peatlands in Kalimantan, Indonesia.

Peatlands in Kalimantan have experienced increasing environmental pressure due to land-use change, drainage, and agricultural expansion (Horton et al., 2021, 2022). Consequently, the region frequently experiences peatland fires, particularly during dry seasons and prolonged drought periods associated with climate variability. Due to the large extent of peatlands and the high occurrence of fire events, Kalimantan provides a suitable and representative study area for investigating tropical peatland fire risk and developing spatio-temporal modeling approaches for fire prediction.

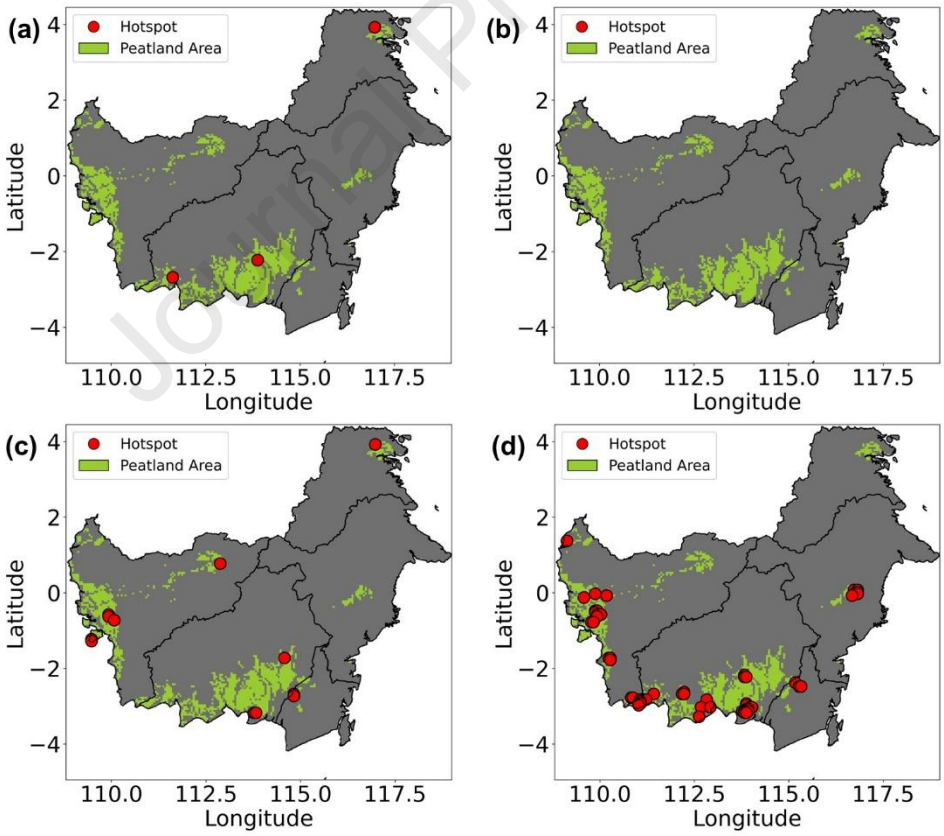
150

151 **2.2. Data**

152 **2.2.1 Data Sources**

153 Fire occurrence in this study was represented by hotspot detections obtained from the
154 Fire Information for Resource Management System (FIRMS). The data were obtained from the
155 Moderate Resolution Imaging Spectroradiometer (MODIS) Collection 6 (Giglio et al., 2016).
156 For the analysis, only hotspot detections with a confidence level greater than 80% were
157 included. The FIRMS dataset provides satellite-based observations of active fires and is used
158 to monitor the spatial and temporal distribution of fire activity. Hotspot data were collected
159 from 1 January 2020 to 31 December 2024 at daily temporal resolution (**Figure 2**).

160



161

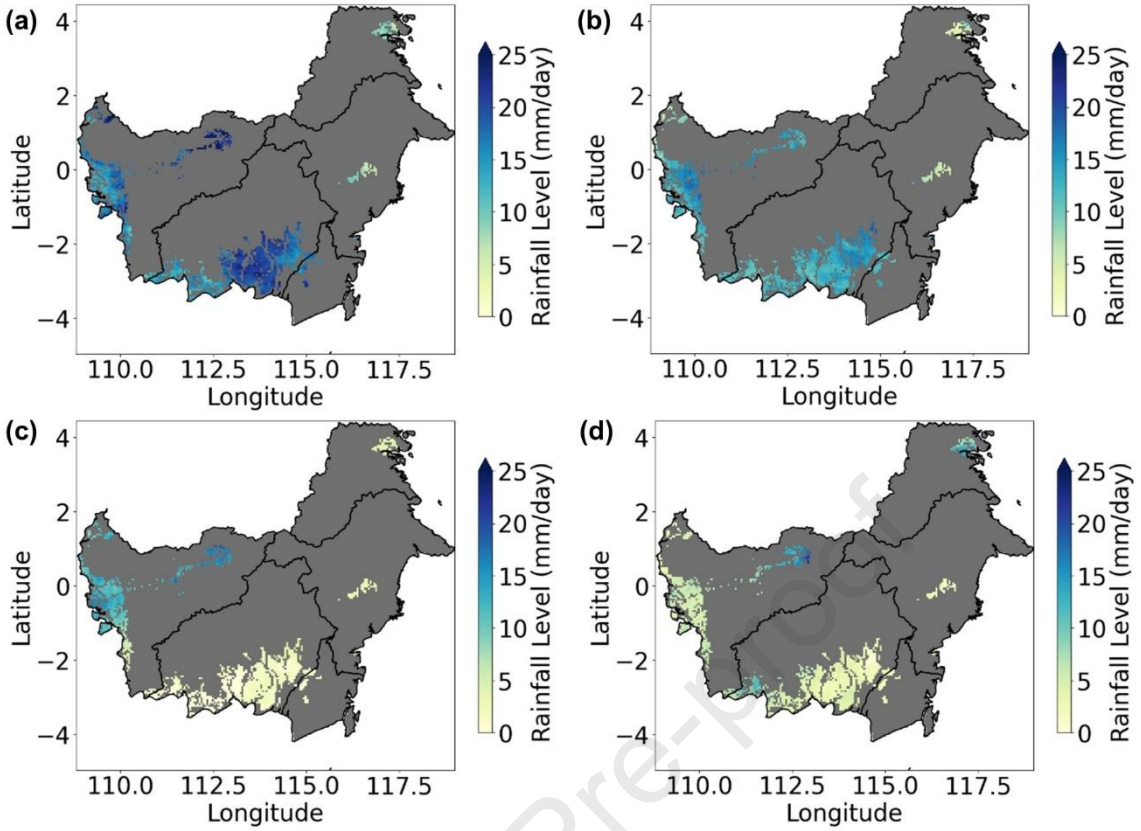
162 **Figure 2. Fire hotspot data snapshots in the study area on (a) 12 January 2024, (b) 8**
163 **March 2024, (c) 16 August 2024, and (d) 20 September 2024.**

Journal Pre-proof

Journal Pre-proof

Meteorological and hydrological conditions were also included to characterize factors influencing tropical peatland fire occurrence. These variables were collected over the same five-year period at daily resolution. Meteorological conditions were represented by rainfall (**Figure 3**) and air temperature (**Figure 4**), which regulate evapotranspiration and influence water table dynamics in peatland ecosystems. Rainfall data were obtained from the Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) (Funk et al., 2015), while air temperature was derived from the ERA5-Land reanalysis dataset produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) (Muñoz-Sabater et al., 2021). Hydrological conditions were represented by soil moisture (**Figure 5**), also obtained from ERA5-Land (Muñoz-Sabater et al., 2021). Soil moisture reflects the water content of the surface peat layer and directly affects fuel availability. Lower soil moisture reduces peat water content, increasing oxygen availability, and facilitating fire ignition and spread. Contrastingly, higher soil moisture maintains saturated conditions that limit oxygen diffusion and reduce the likelihood of sustained combustion.

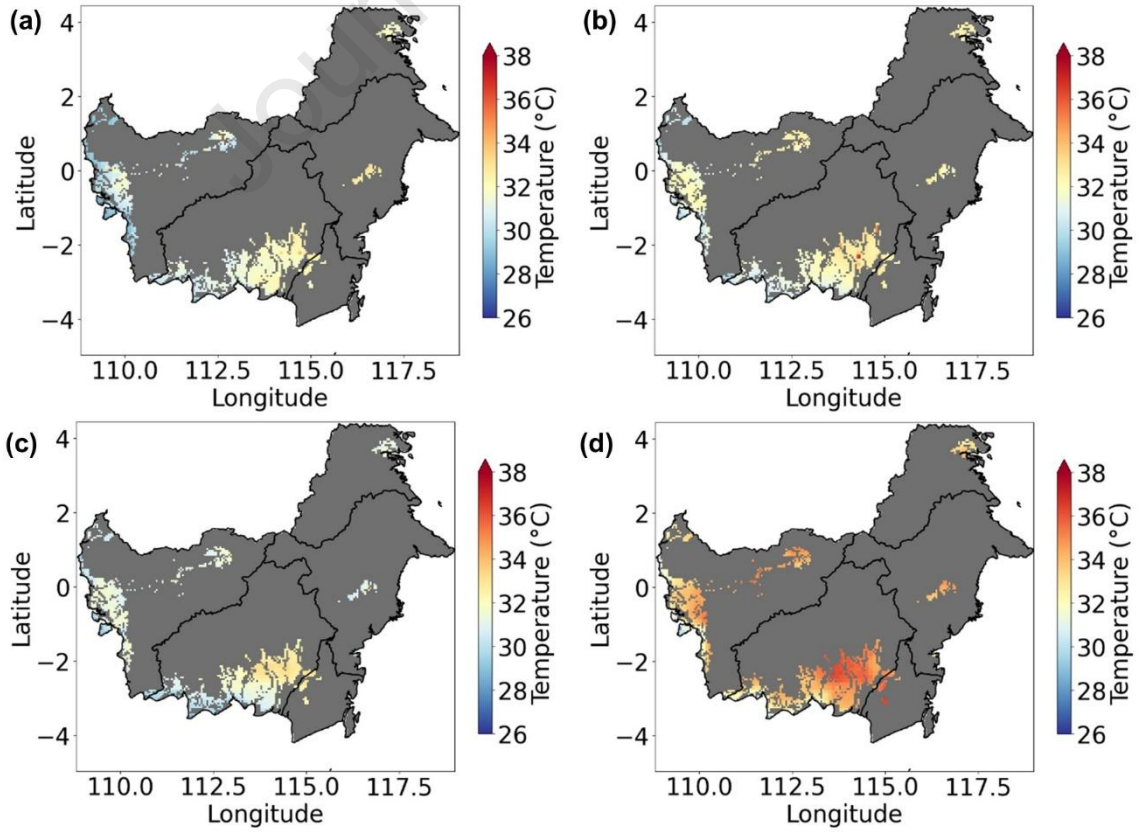
Journal Pre-proof



178

179 **Figure 3. Rainfall data snapshots in the study area on (a) 12 January 2024, (b) 8 March**

180 **2024, (c) 16 August 2024, and (d) 20 September 2024.**

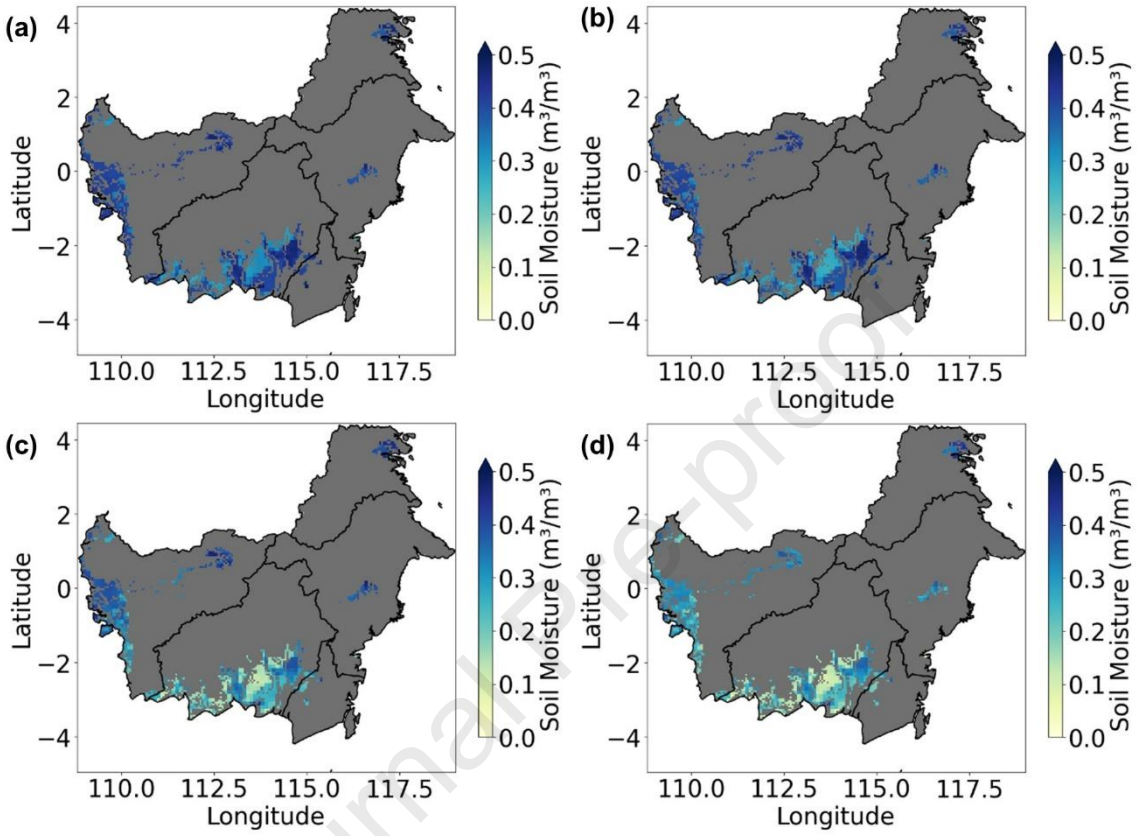


181

Journal Pre-proof

182 **Figure 4. Temperature data snapshots in the study area on (a) 12 January 2024, (b) 8**
183 **March 2024, (c) 16 August 2024, and (d) 20 September 2024.**

184



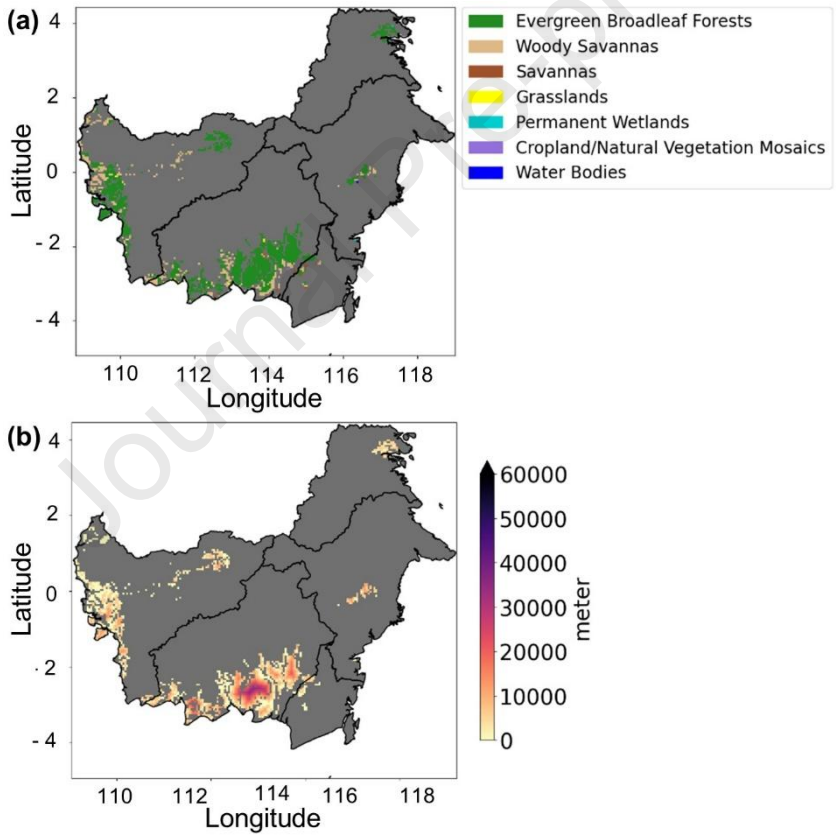
185

186 **Figure 5. Soil moisture data snapshots in the study area on (a) 12 January 2024, (b) 8**
187 **March 2024, (c) 16 August 2024, and (d) 20 September 2024.**

188

189 In addition to meteorological and hydrological variables, the dataset also incorporates
190 land cover and distance to human activities (**Figure 6**) to represent the spatial characteristics
191 of the landscape and human influences on fire occurrence. Land cover data were obtained from
192 the MODIS Collection 6 Land Cover Type Product (Sulla-Menashe et al., 2019). This dataset
193 characterizes the dominant vegetation or land use within each spatial unit, which determines
194 the structure of available fuels, biomass accumulation, and baseline flammability conditions.
195 Distance to human activities was derived from the Global Human Settlement Layer (GHSL)

196 developed by the European Commission Joint Research Centre (Pesaresi et al., 2024). This
197 variable represents the proximity of each location to areas associated with human presence and
198 is used as a proxy for anthropogenic influence on fire ignition. In tropical peatland regions,
199 fires are often associated with land clearing, agricultural expansion, and other land
200 management practices. As a consequence, proximity to human activities is an important
201 variable influencing fire occurrence. **Table 1** summarizes the data and corresponding sources
202 employed in this study, which provide the foundation for constructing the input variables for
203 model development.
204



205

206 **Figure 6. Spatial distribution of (a) land cover and (b) distance to human activities. Land**
207 **cover indicates vegetation type and potential fuel availability across peatland areas.**
208 **Distance to human activities represents anthropogenic pressure and potential fire ignition**
209 **sources.**

Journal Pre-proof

Table 1. Summary of datasets and corresponding sources employed in this study.

Data	Source	Reference
Peatland distribution	Geospatial Information Agency of Indonesia (BIG)	Geospatial Information Agency (2019)
Hotspot	Fire Information for Resource Management System (FIRMS)	Giglio et al. (2016)
Rainfall	Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS)	Funk et al. (2015)
Air temperature	ERA5-Land reanalysis dataset	Muñoz-Sabater et al.(2021)
Soil moisture	ERA5-Land reanalysis dataset	Muñoz-Sabater et al.(2021)
Land cover classification	MODIS Collection 6 Land Cover Type Product	Sulla-Menashe et al.(2019)
Distance to human activity	Global Human Settlement Layer (GHSL)	Pesaresi et al.(2024)

2.2.2 Data Preprocessing

Prior to modeling, the datasets were preprocessed to ensure spatial consistency and compatibility for machine learning analysis. Because the data were obtained from multiple sources with different spatial resolutions and formats, all variables were resampled to a uniform spatial grid with a resolution of 5 × 5 km. This resolution was selected to align all variables within the same spatial reference system and to minimize potential biases arising from

differences in spatial scale. The use of a 5 × 5 km grid is also consistent with previous grid based fire analyses. For example, Tan et al. (2020) resampled multi source fire related datasets to a 5 km grid in an analysis of forest and land fires in Indonesia, noting that this resolution represented a trade-off between spatial accuracy and computational feasibility. Similar 5 × 5 km grid structures have been used to summarize fire counts and model fire occurrence in the Western Himalayas (Bar et al., 2023), and to analyze hotspot density and dry lightning wildfire probability in Tasmania (Wickramasinghe et al., 2024). However, the 5 × 5 km grid should be interpreted as a regional scale fire occurrence probability unit rather than a fine scale fire patch detection resolution.

To quantify the spatial dependence of the continuous input variables, global Moran's I was calculated after all datasets were resampled to the 5 × 5 km spatial grid. The analysis was applied to rainfall, air temperature, soil moisture, and distance to human activities, which are represented as continuous gridded predictors. For the time-varying variables, including rainfall, air temperature, and soil moisture, Moran's I was calculated from daily spatial maps, and the annual average Moran's I and percentage of statistically significant daily maps were summarized. The results show consistently positive spatial autocorrelation across all variables. Rainfall showed the highest spatial dependence, with annual average Moran's I values ranging from 0.910 to 0.934. Air temperature also showed strong spatial autocorrelation, with annual average Moran's I values ranging from 0.815 to 0.875. Soil moisture showed positive spatial autocorrelation, with annual average Moran's I values ranging from 0.742 to 0.780. In all years, 100% of the daily maps for these three variables were statistically significant. Distance to human activities also exhibited strong positive spatial autocorrelation, with a Moran's I value of 0.800. These results indicate that the continuous input layers contain strong spatially structured information, supporting the use of CNN-based convolutional filters to extract local spatial features from neighboring grid cells.

Before normalization, the stationarity of the time-varying continuous input variables was evaluated using the Augmented Dickey-Fuller (ADF) test. The test was applied to rainfall, air temperature, and soil moisture to assess whether these variables contained a unit root. The null hypothesis of the ADF test is that the time series is non-stationary. The results showed that the null hypothesis was rejected for all three variables at the 5% significance level. Rainfall had an ADF statistic of -4.5865 with a p-value of 0.0001, air temperature had an ADF statistic of -4.0835 with a p-value of 0.0010, and soil moisture had an ADF statistic of -4.2116 with a p-value of 0.0006. These results indicate that rainfall, air temperature, and soil moisture were stationary before normalization.

Continuous variables, including rainfall, air temperature, soil moisture, and distance to human activity, were normalized to the range [0, 1] to ensure comparable numerical scales and prevent bias toward features with larger magnitudes during model training (Horton et al., 2021). The normalization is expressed as

$$x_i^* = \frac{x_i - x_{min}}{x_{max} - x_{min}} \quad (1)$$

where x_i is the original value, $x_{\min}$ and $x_{\max}$ are the minimum and maximum values of the corresponding variable, and x_i^* is the transformed value after normalization. Land cover data, originally represented as categorical integer labels, were transformed using one-hot encoding to ensure that each class was treated as an independent nominal category rather than an ordinal value. For the target variable, a grid cell was labeled as a fire if at least one hotspot detection was recorded within the cell. Conversely, grid cells without any detected fire events were labeled as non-fire. This binary classification approach enables the model to learn spatial and temporal patterns associated with fire occurrence across the study area.

To address the challenge of class imbalance between fire and non-fire classes, a weighted loss function was employed (Y. Ho & S. Wookey, 2020) during data preprocessing

$$L_w = -\sum_{i=1}^{n}\left(w\, y_i \log(\hat{y}_i) + \left(1 - y_{i,}\right)\log(1 - \hat{y}_i)\right) \quad (2)$$

where L_w is the weighted loss function, n is the number of samples, y_i is the label for i^{th} sample, $\hat{y}_i$ is the predicted probability of fire occurrence produced by the model, w is the positive constant weight coefficient. Non-fire observations typically dominate the dataset which can bias the model toward predicting the majority class. The weighted loss function assigns higher penalty weights to the minority class, enabling the model to better learn patterns associated with fire occurrences and improving its sensitivity to rare fire events.

2.3 Model

The proposed modeling framework is designed to capture the complex spatio–temporal dynamics of tropical peatland fire occurrence by integrating machine learning architectures with a probabilistic calibration method. The framework combines Convolutional Neural Networks (CNN) for spatial feature extraction, Convolutional Long Short-Term Memory (ConvLSTM) for learning temporal dependencies, and isotonic regression for probability calibration. This integrated approach enables the model to represent the spatio-temporal evolution of fire risk and produce calibrated probabilistic predictions suitable for risk analysis (**Figure 7**).

2.3.1 Model formulation

CNN is employed to extract spatial features from tropical peatland fire predictor maps at each observation time (Khirani et al., 2026; Yar et al., 2023; Zelioli et al., 2025). The main operation in CNN is the convolution process, which is defined as

$$F_t(i,j) = \sum_{m}\sum_{n} X_i\,(i-m, j-n)\,K(m,n) \quad (3)$$

Journal Pre-proof

where $F_t(i,j)$ is the resulting feature value at spatial position (i,j), X_t is the input of spatial grid, and K is the convolution kernel. After convolution process, a nonlinear activation function is introduced to enhance the model capacity through the Rectified Linear Unit (ReLU)

$$\mathrm{ReLU}(x) = \max(0, x) \tag{4}$$

Journal Pre-proof

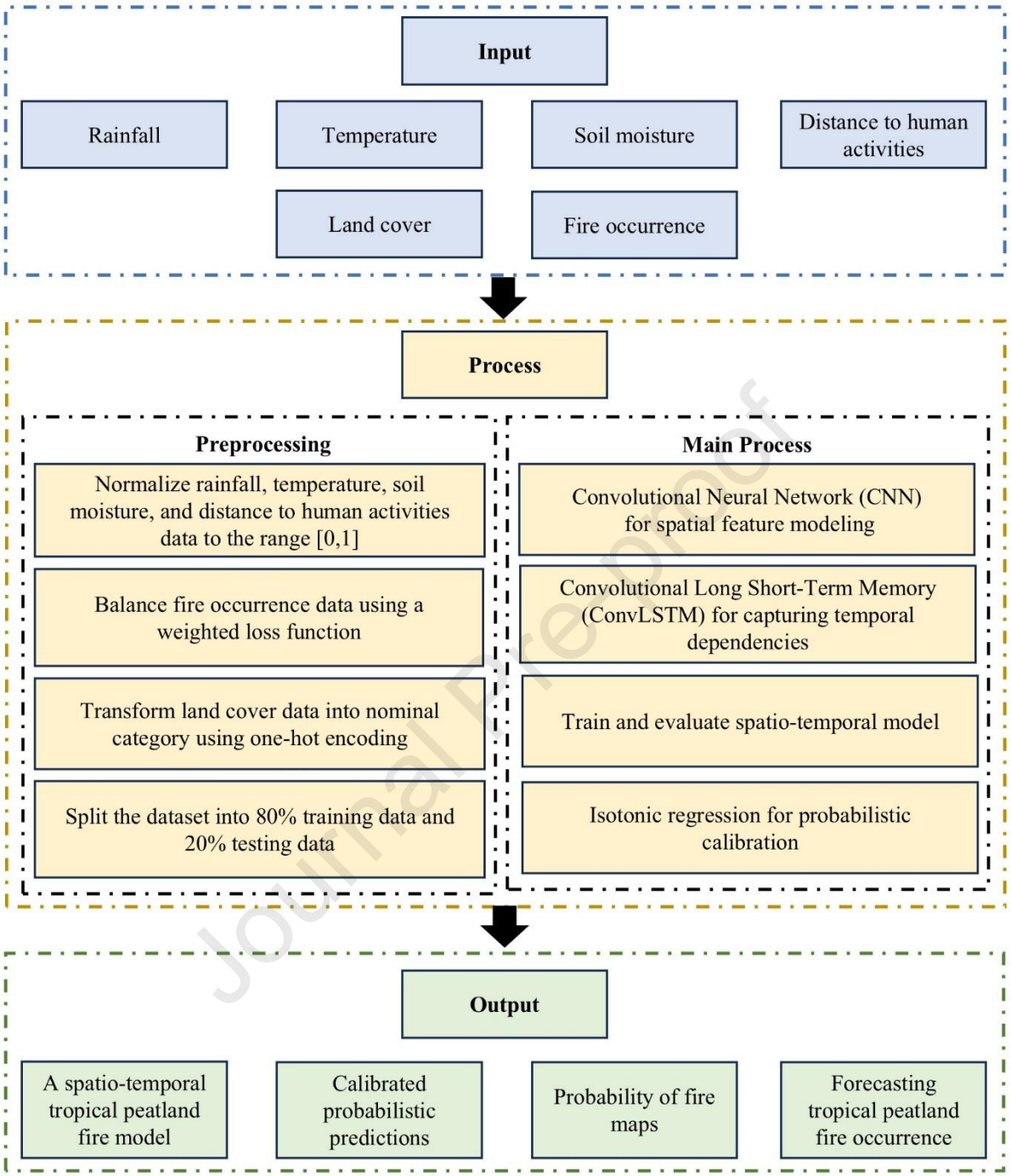


Figure 7. Conceptual diagram of the proposed model consisting of input, process, and output stages. The input stage includes meteorological, hydrological, land cover, anthropogenic, and fire occurrence variables. The process stage involves preprocessing data and main modeling process. Preprocessing data is conducted through normalization, class balancing using a weighted loss function, one-hot encoding of land cover data, and

training–testing data partitioning. The main modeling process integrates CNN for spatial feature extraction, ConvLSTM for temporal dependency learning, and isotonic regression for probabilistic calibration. The output stage produces calibrated probability maps for analyzing and forecasting tropical peatland fire, which support early warning, risk assessment, and fire mitigation efforts

The spatial feature maps extracted from several consecutive time steps are arranged into a spatio-temporal sequence and passed to a ConvLSTM layer, which extends the standard LSTM architecture by replacing matrix multiplication with convolution (Khalaf et al., 2024; Mambile et al., 2025; Xu et al., 2025). Consequently, it can preserve spatial information and capture temporal dependencies. The ConvLSTM equations are written as

$$i_t = \sigma(W_{xi} * F_t + W_{hi} * H_{t-1} + b_i) \quad (5)$$

$$f_t = \sigma(W_{xf} * F_t + W_{hf} * H_{t-1} + b_f) \quad (6)$$

$$o_t = \sigma(W_{xo} * F_t + W_{ho} * H_{t-1} + b_o) \quad (7)$$

$$\tilde{C}_t = \tanh(W_{xc} * F_t + W_{hc} * H_{t-1} + b_c) \quad (8)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (9)$$

$$H_t = o_t \odot \tanh(C_t) \quad (10)$$

where σ sigmoid activation function, $*$ is the convolution operator, $\odot$ is the element-wise multiplication, i_t is the input gate, f_t is the forget gate, o_t is the output gate, $W_{xi}, W_{xf}, W_{xo}, W_{xc}$, $W_{hi}, W_{hf}, W_{ho}, W_{hc}$ are the trainable convolution weights, b_i, b_f, b_o, b_c are the bias terms, F_t is the spatial feature map, $\tilde{C}_t$ is the candidate cell state, $C_t$is the updated cell state, and H_t is the hidden state. The final hidden state is mapped to the predicted fire risk through an output convolution layer

$$\hat{R}_t = g(H_t) \quad (11)$$

Journal Pre-proof

where $\hat{R}_t$ denotes the predicted spatial distribution of tropical peatland fire risk and g is the output mapping function.

To improve the reliability of the predicted outputs, isotonic regression (Delong & V. Wüthrich, 2025; Gneiting & Resin, 2023) is applied as a post processing calibration step. Although the CNN–ConvLSTM model can effectively capture relative fire susceptibility, its raw outputs may not accurately correspond to the true probability of fire occurrence. Isotonic regression addresses this issue by learning a monotonic mapping from the raw prediction to a calibrated probability

$$p_k = m(\hat{R}_k) \tag{12}$$

with the monotonicity constraint

$$\hat{R}_k \leq \hat{R}_l \Rightarrow m(\hat{R}_k) \leq m(\hat{R}_l) \tag{13}$$

where p_k is the calibrated probability, m is the calibration function, and $\hat{R}_k$ is the raw prediction from the CNN–ConvLSTM model. The calibration function is estimated by minimizing the squared error between observed outcomes and calibrated values

$$\min_m \sum_{k=1}^{N} \left(Y_k - m(\hat{R}_k)\right)^2 \tag{14}$$

where N is the number of data points in the calibration dataset and Y_k is the observed fire occurrence indicator. This calibration preserves the rank ordering of the original predictions and improves agreement between predicted values and observed event frequencies. Therefore, the proposed framework provides both spatio-temporal model of tropical peatland fire risk dynamics and calibrated probabilistic outputs suitable for risk assessment and early warning applications.

2.3.2 Model implementation

Journal Pre-proof

Journal Pre-proof

The prepared data were partitioned into training and testing sets using a chronological split to preserve temporal ordering. The dataset covered the period from 1 January 2020 to 31 December 2024. The first 80% of the observations, corresponding to 1 January 2020 to 31 December 2023, were used for model training, while the last 20%, corresponding to 1 January 2024 to 31 December 2024, were reserved for testing. This chronological partition ensures that the model is evaluated on unseen future data and reduces the risk of information leakage from later observations into the training process.

The CNN–ConvLSTM architecture consisted of four main layers. The first layer was a ConvLSTM2D layer employed as the spatio-temporal feature extractor, with 32 filters, a 3 × 3 kernel size, tanh activation, and sigmoid recurrent activation. Batch normalization and dropout with a rate of 0.30 were applied after this layer to stabilize training and reduce overfitting. The second layer was a Conv2D decoder layer for extracting deeper spatial features, with 64 filters, a 3 × 3 kernel size, and ReLU activation, followed by batch normalization and dropout with a rate of 0.30. The third layer was a Conv2D feature summary layer with 32 filters, a 3 × 3 kernel size, and ReLU activation, followed by batch normalization. The final layer was a Conv2D output layer with one filter and a 1 × 1 kernel size. A sigmoid activation function was used in the output layer to produce one probability map with values constrained between 0 and 1. The model was trained using a batch size of 2. During the initial hyperparameter search, the maximum number of epochs was set to 70, with early stopping applied using a patience of 20 epochs. For the deeper training search, early stopping was also applied, with a patience of 30 epochs. These early stopping criteria were used to prevent overfitting and to terminate training when the validation performance no longer improved.

Because fire events were relatively rare compared with non-fire observations, the training data exhibited substantial class imbalance. To address this issue, a positive class weight was applied during model training. Several candidate positive class weights, including 10, 25,

50, 75, and 100, were evaluated using the testing period to examine the precision–recall trade off. Based on this evaluation, a positive class weight of 75 was selected because it provided the most balanced detection performance, with precision of 0.699 and recall of 0.885 (**Table 2**). This setting was therefore used in the final model to improve sensitivity to fire events while maintaining acceptable precision

Table 2. Precision and recall values obtained under different positive class weights. The positive class weight of 75 was selected because it provided the best balance between precision and recall for detecting fire occurrence in the testing period.

Positive class weight	Precision	Recall
10	0.719	0.025
25	0.464	0.703
50	0.389	0.325
75	0.699	0.885
100	0.459	0.793

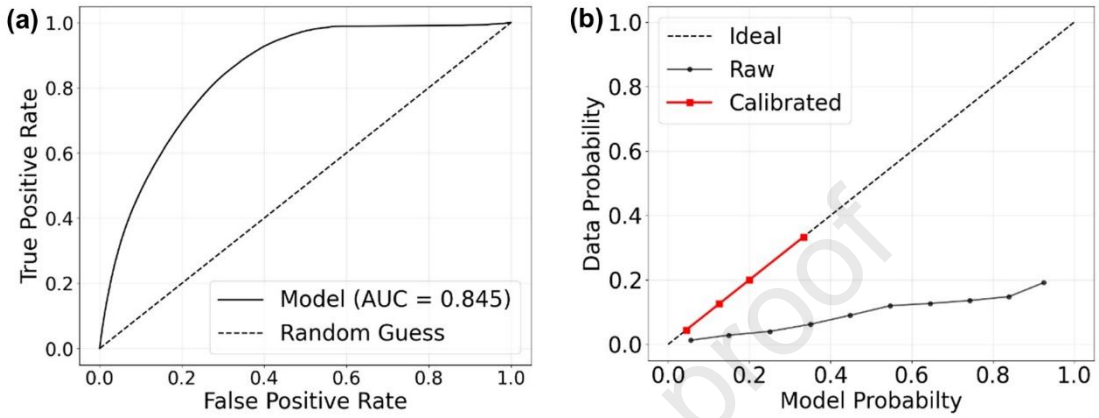
3. Results

3.1 Model performance

We assessed model performance by examining both discrimination ability and probability reliability. The Receiver Operating Characteristics (ROC) curve shows that the proposed model achieves an Area Under the Curve (AUC) of 0.845, which indicates good classification performance in distinguishing fire occurrence from non-fire conditions (**Figure 8a**). Model performance is further evaluated through calibration analysis. The raw probability outputs from the model are not properly calibrated, because the calibration curve falls below the ideal diagonal line with the value of Brier score is around 0.198. This pattern indicates a discrepancy between the predicted probabilities and the observed event frequencies. Following

Journal Pre-proof

379 isotonic regression, the calibrated curve shifts closer to the ideal relationship, particularly over
380 the probability range where the majority of predictions are distributed with the value of the
381 Brier score decreased substantially to approximately 0.011 (**Figure 8b**).
382



383

384 **Figure 8. Model performance evaluation based on (a) the ROC–AUC curve and (b) the**
385 **probability curve. The ROC–AUC curve shows the discriminatory ability of the model,**
386 **while the probability curve examines the consistency between predicted probabilities and**
387 **observed fire occurrence frequencies.**
388

389 To evaluate whether model performance was robust under contrasting seasonal
390 conditions, the testing period was divided into dry and wet seasons. The dry season was defined
391 as May–October, while the wet season was defined as November–April. For each season,
392 ROC–AUC was calculated to assess the discrimination ability of the model, while the Brier
393 score was used to evaluate the reliability of the predicted probabilities.

394 The seasonal evaluation shows that the model maintained stable discrimination
395 performance across both seasons. The ROC–AUC values were 0.811 for the dry season and
396 0.822 for the wet season, which are broadly consistent with the overall ROC–AUC of 0.845
397 obtained for the full testing period. This indicates that the model was able to distinguish fire
398 and non-fire conditions under both fire prone dry conditions and wetter periods with lower fire

Journal Pre-proof

activity. The Brier scores for the dry and wet seasons were 0.088 and 0.027, respectively. These values are also consistent with the overall Brier score of around 0.011 for the full testing period. The higher Brier score during the dry season suggests greater probabilistic uncertainty under more fire prone conditions, whereas the lower wet season Brier score indicates relatively more reliable probability estimates during periods of reduced fire occurrence. These seasonal results support the robustness of the proposed model across contrasting hydroclimatic conditions.

The probability of fire results (**Figure 9**) indicate that the proposed spatio-temporal modeling framework is able to capture the main patterns of tropical peatland fire occurrence in Kalimantan. The spatial distribution of predicted fire probability indicates that fire risk is not uniformly distributed across the study area but is concentrated in peatland areas associated with woody savanna land cover and short distances to human activities. Furthermore, the model also captures the temporal variability of fire activity, with higher probabilities during the drier months of August and September, when peatland conditions become more vulnerable to ignition and fire spread.

To further evaluate the temporal robustness of the model, a rolling origin validation strategy was conducted in addition to the main chronological 80:20 train-test split (Farghaly et al., 2026; Fiorucci & Louzada, 2020). This validation strategy preserves the temporal structure of the data and avoids information leakage from future observations into the training period. Three rolling folds were evaluated. In Fold 1, data from 2020–2021 were used for training and data from 2022 were used for validation. In Fold 2, data from 2020–2022 were used for training and data from 2023 were used for validation. In Fold 3, data from 2020–2023 were used for training and data from 2024 were used for validation.

The rolling validation results show that the model generally maintained acceptable discrimination performance across different validation years. The ROC–AUC values were 0.8622 for Fold 1, 0.6987 for Fold 2, and 0.8538 for Fold 3. The minimum and maximum

Journal Pre-proof

ROC–AUC values across repeated runs were 0.8520–0.8784 for Fold 1, 0.6844–0.7211 for Fold 2, and 0.7211–0.8639 for Fold 3. The corresponding Brier scores were 0.0872, 0.1891, and 0.1729, respectively. The lower ROC–AUC and higher Brier score in Fold 2 indicate that the 2023 validation period was more difficult to predict, suggesting stronger interannual variability in fire risk conditions. The high ROC–AUC values in Folds 1 and 3 indicate that the proposed model can capture temporally transferable fire risk patterns. The rolling validation supports the robustness of the proposed framework while highlighting the need to account for interannual variability in operational fire forecasting.

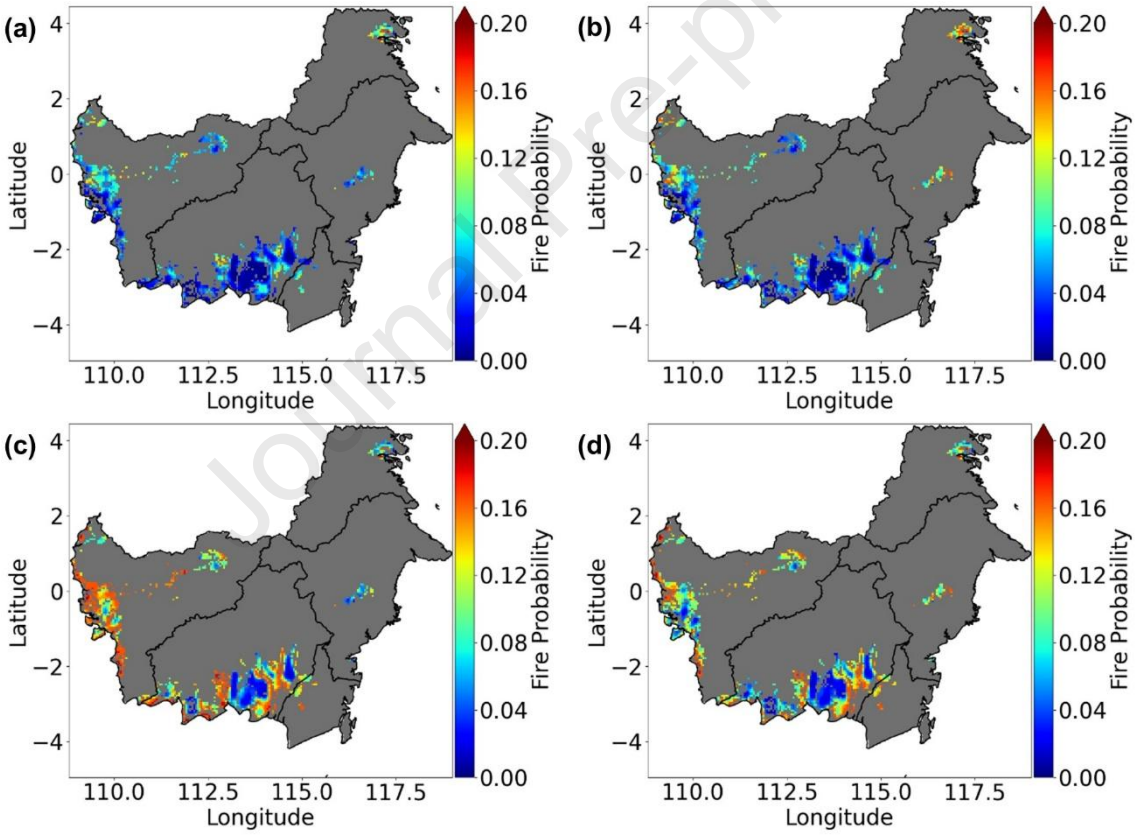


Figure 9. Probability maps of tropical peatland fire risk on (a) 12 January 2024, (b) 8 March 2024, (c) 16 August 2024, and (d) 20 September 2024.

Journal Pre-proof

The uncertainty analysis based on Monte Carlo dropout indicates that the model produced relatively narrow prediction intervals for areas with consistently low fire probability and wider intervals in transition zones between low and elevated fire risk (**Figure 10**). Across the testing period, the average width of the 95% prediction interval was 0.041 for low-risk grid cells, 0.083 for moderate-risk grid cells, and 0.112 for high-risk grid cells. This pattern suggests that prediction uncertainty increases in areas where environmental and hydrological conditions are closer to fire-prone thresholds.

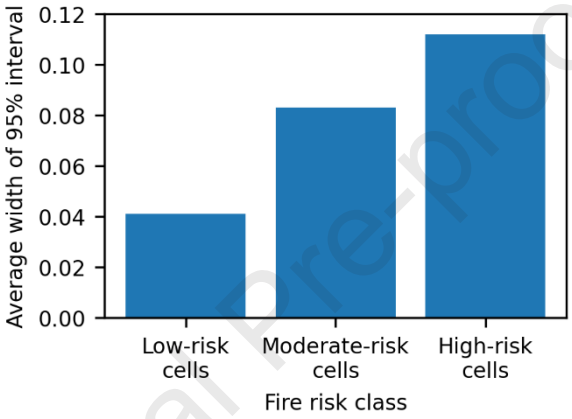


Figure 10. Monte Carlo dropout based uncertainty across fire risk classes. The average width of the 95% prediction interval increased from low-risk grid cells to high-risk grid cells, indicating greater predictive uncertainty in areas with elevated fire probability.

3.2 Forecasting capability

The forecasting capability of the proposed spatio-temporal modeling framework is evaluated by comparing the predicted probability of fire maps for the next three and ten days with the observed hotspots (**Figure 11**). In terms of spatial prediction, the forecast fire risk patterns are in agreement with the actual hotspot occurrences, because the observed hotspots are located within or adjacent to the elevated risk zones identified by the model. In the next three-day forecast, the observed hotspot was located in an area with a mean predicted fire probability of 0.284 and a 95% prediction interval of 0.221–0.336 (**Figure 12**). This indicates

that the area was consistently classified as having elevated fire probability across repeated stochastic predictions. In the next ten-day forecast, the observed hotspot was located in an area with a mean predicted fire probability of 0.247 and a 95% prediction interval of 0.176–0.319. Although the uncertainty interval was wider for the ten-day forecast, the hotspot remained within or adjacent to the elevated-risk zone. These results show that the model can identify fire prone areas while also quantifying the uncertainty associated with the forecasted probability maps.

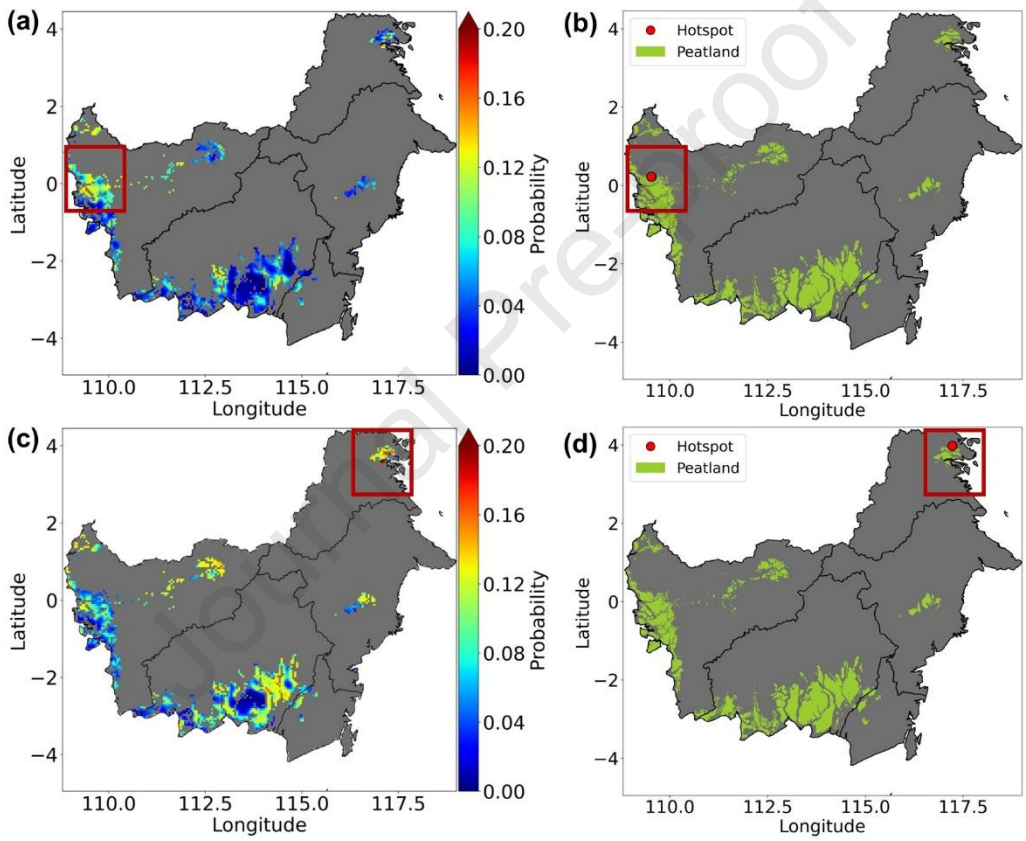


Figure 11. Comparison between forecasted fire probability from the proposed model and actual fire hotspots for (a, b) the next three-day forecast and (c, d) the next ten-day forecast. The red box highlights areas where high predicted fire probability aligns with observed fire hotspots.

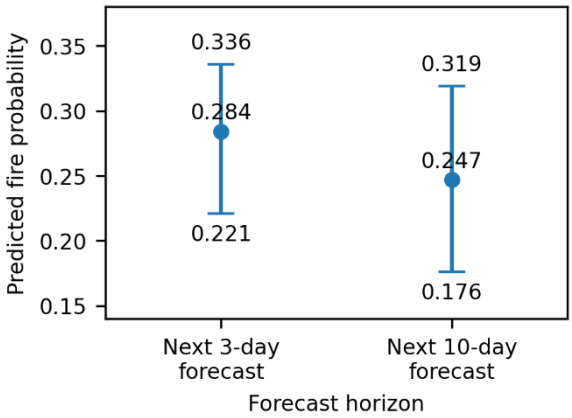


Figure 12. Forecasted fire probability with 95% prediction intervals for observed hotspot areas. The next ten-day forecast showed a wider prediction interval than the next three-day forecast, indicating greater uncertainty at the longer forecast horizon.

The forecasts also show that the fire risk distribution changes across the two prediction horizons. For the next three-day forecast, the model identifies an area of increased fire probability in the western peatland region, with the observed hotspot located within the predicted high risk area. For the next ten-day forecast, the predicted fire risk distribution shifts and emphasizes elevated probabilities in the northern region, where the observed hotspot is also recorded. These results suggest that the model is able to capture the evolution of fire prone conditions over time.

To complement the visual comparison between predicted fire probability maps and observed hotspots, quantitative validation was conducted for both forecast horizons. The evaluation was performed at the grid-cell level by comparing the predicted fire probabilities with the observed fire occurrence labels during the corresponding forecast periods. Three metrics were used: ROC–AUC, precision, and recall. The next three-day forecast achieved a ROC–AUC of 0.836, precision of 0.688, and recall of 0.842. These values indicate good short-term discrimination ability and show that most observed fire occurrences were captured within predicted elevated risk areas. For the next ten-day forecast, the ROC–AUC decreased to 0.784,

Journal Pre-proof

with precision of 0.670 and recall of 0.764. The lower performance for the ten-day horizon reflects the greater uncertainty associated with longer forecast periods.

3.3 Variable importance analysis

To improve model interpretability, permutation importance analysis was conducted to evaluate the contribution of each predictor to fire occurrence prediction (Hapfelmeier et al., 2023; Mehdiyev et al., 2025). Because the input variables consisted of both temporal and non-temporal spatial predictors, two permutation strategies were used. For temporal variables, including rainfall, air temperature, and soil moisture, the temporal order was randomized while the remaining variables were kept unchanged. For non-temporal spatial variables, including distance to human interaction and land-cover type, the spatial locations were randomized to evaluate the importance of their spatial structure.

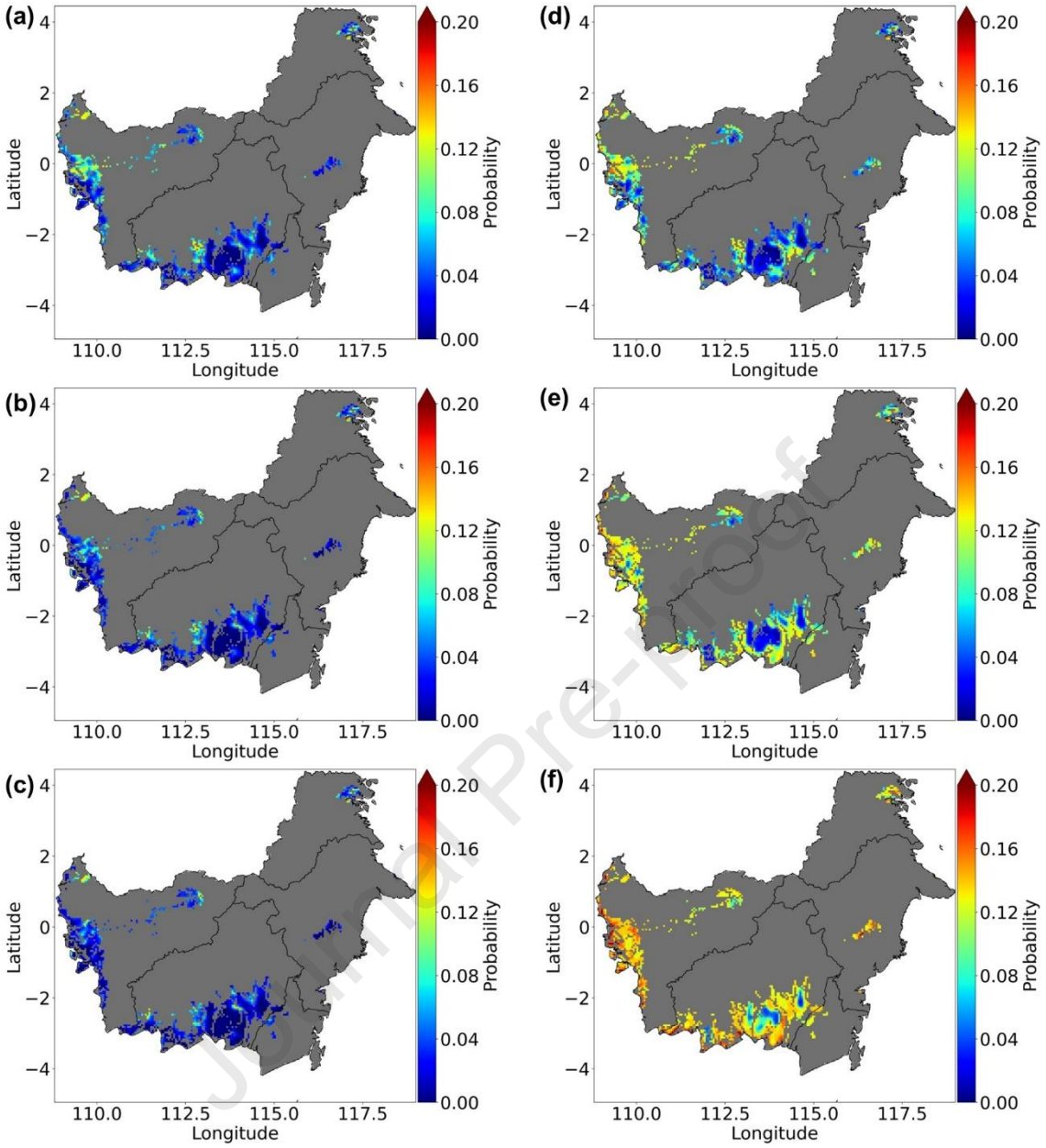
The permutation importance results show that land cover type and rainfall were the most influential variables in the model. Randomizing land cover type produced the largest decrease in performance, with reductions of 0.085 in ROC–AUC. This indicates that spatial differences in land cover provide important information for distinguishing fire-prone and non-fire-prone areas. Among the temporal predictors, rainfall showed the strongest contribution, with decreases of 0.060 in ROC–AUC after temporal permutation. Distance to human interaction also had an important contribution, with decreases of 0.057 in ROC–AUC, indicating the relevance of anthropogenic proximity in explaining fire occurrence. Soil moisture and air temperature also contributed to model performance, although their effects were smaller.

3.4 Sensitivity analysis

Journal Pre-proof

We conduct sensitivity analysis for the next three-day (**Figure 13**) and the next ten-day (**Figure 14**) forecasts to assess the response of the predicted fire probability to controlled perturbations in the rainfall, air temperature, and soil moisture. Two contrasting scenarios are examined using perturbation levels of 10%, 30%, and 50%. These perturbations were applied relative to the original input values used for each forecast horizon. Therefore, the unperturbed next three-day and next ten-day forecast inputs were used as the reference scenarios, rather than historical mean values. The first scenario characterized by increased rainfall, increased soil moisture, and decreased air temperature, representing a wetter and cooler environmental state. In contrast, the second scenario represented by decreased rainfall, decreased soil moisture, and increased air temperature, representing a drier and warmer environmental state associated with lower water availability and enhanced evapotranspiration. These perturbations are designed to evaluate the model response under varying meteorological and hydrological conditions.

The sensitivity results demonstrate that the model exhibits a consistent response across both forecasting horizons, including the next three-day and the next ten-day. Under the first scenario, the predicted probability of fire decreases progressively with increasing perturbation, and the extent of susceptible areas contracts across the study region. Under the second scenario, the predicted probability of fire increases substantially, and the spatial extent of elevated fire risk areas expands with increasing perturbation magnitude from 10% to 50%.



535

536 **Figure 13. Sensitivity analysis for the next three-day forecast under the first scenario (a,**
537 **b, c) and the second scenario (d, e, f) with perturbation levels of 10%, 30%, and 50%,**
538 **respectively. The first scenario represents wetter and cooler conditions, with increased**
539 **rainfall, increased soil moisture, and decreased air temperature. Contrastingly, the**
540 **second scenario represents drier and warmer conditions, with decreased rainfall,**
541 **decreased soil moisture, and increased air temperature.**

542

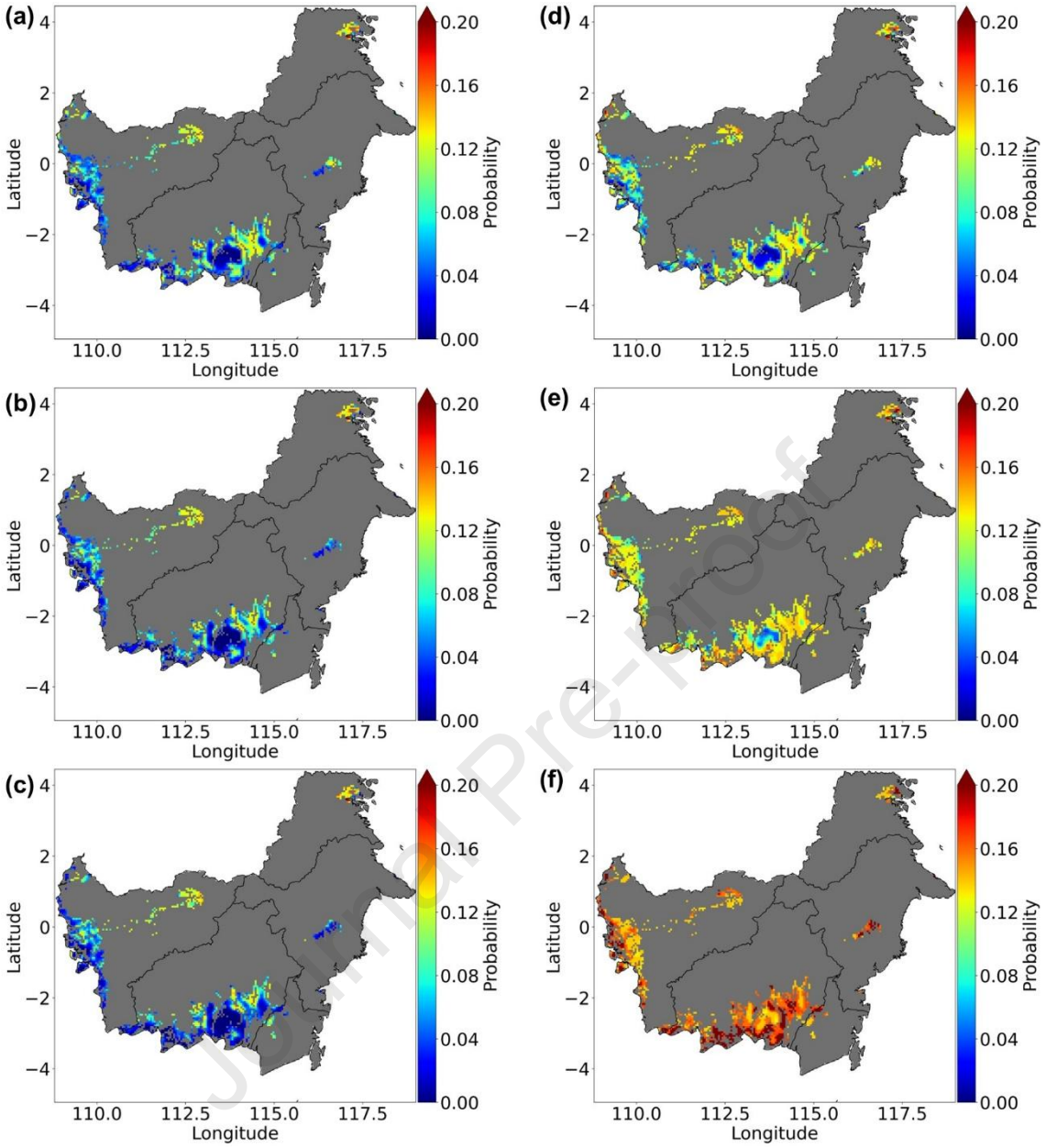


Figure 14. Sensitivity analysis for the next ten-day forecast under the first scenario (a, b, c) and the second scenario (d, e, f) with perturbation levels of 10%, 30%, and 50%, respectively. The first scenario represents wetter and cooler conditions, with increased rainfall, increased soil moisture, and decreased air temperature. Contrastingly, the second scenario represents drier and warmer conditions, with decreased rainfall, decreased soil moisture, and increased air temperature.

4. Discussion

552 The proposed model integrates Convolutional Neural Networks (CNN) (Khirani et al.,
553 2026; Yar et al., 2023; Zelioli et al., 2025), Convolutional Long Short-Term Memory
554 (ConvLSTM) (Khalaf et al., 2024; Mambile et al., 2025; Xu et al., 2025), and isotonic
555 regression (Delong & V. Wüthrich, 2025; Gneiting & Resin, 2023) to provide a unified
556 representation of the spatial, temporal, and probabilistic characteristics of tropical peatland fire.
557 The CNN component extracts spatially structured features from gridded environmental
558 predictors, allowing the model to learn patterns associated with fire prone conditions. These
559 spatial features are subsequently processed by the ConvLSTM component, which captures
560 temporal evolution while preserving spatial dependence. Consequently, the model
561 appropriately represents the development, persistence, and shift of fire susceptibility. The
562 inclusion of isotonic regression further improves the framework by calibrating the predicted
563 probabilities and increasing the consistency of the model outputs with the observed frequency
564 of fire events.

565 In comparison with existing approaches for assessing tropical peatland fire risk,
566 including the Peat Fire Vulnerability Index (PFVI) (Taufik et al., 2022) and machine learning
567 models (Horton et al., 2021; Mahdiyasa et al., 2025), the proposed model provides a more
568 comprehensive representation of fire dynamics (**Table 3**). PFVI provides simple and physically
569 interpretable measures of fire danger under drying conditions, while machine learning models
570 offer greater flexibility for modeling nonlinear relationships among variables. However, these
571 approaches often capture only certain aspects of tropical peatland fire behaviour, either through
572 index-based representations of vulnerability or through models that focus primarily on spatial
573 or temporal patterns separately. The proposed model extends these capabilities by integrating
574 spatial feature extraction, temporal dependence, and probability calibration within a single
575 framework.

Journal Pre-proof

576 **Table 3.** Comparison of the proposed model, the spatial machine learning model of Horton et
577 al. (2021) the temporal machine learning model of Mahdiyasa et al. (2025) and the Peat Fire
578 Vulnerability Index (PFVI) of Taufik et al. (2022) for analyzing tropical peatland fire dynamics.

	Spatial dependence	**Temporal dependence**	**Fire risk**
Proposed model	Explicitly represented through CNN-based spatial feature extraction	Explicitly represented through ConvLSTM-based temporal learning	Calibrated probability of fire through isotonic regression
Spatial machine learning model from Horton et al. (2021)	Explicitly represented through XGBoost and Random Forest methods	Not incorporated in the model	Uncalibrated predicted probability
Temporal machine learning model from Mahdiyasa et al. (2025)	Not incorporated in the model	Explicitly represented through LSTM and GRU methods	Fire index or fire danger class
Peat Fire Vulnerability Index from Taufik et al. (2022)	Not incorporated in the model	Explicitly represented through empirical relationship	Fire index or fire danger class

Journal Pre-proof

Journal Pre-proof

In addition to empirical indices and machine learning models, the integration of IoT-based monitoring systems and the Fire Weather Index (FWI) has also been widely discussed for peatland fire monitoring and early warning. A recent study highlighted the integration of IoT-based environmental monitoring and FWI systems for improving fire prediction and mitigation (Li, 2025). Other studies have shown that IoT-enabled systems combined with neural networks can predict groundwater level using atmospheric parameters with root mean square error values between 3.554 and 4.920 mm, supporting real time fire risk mitigation in peatlands (Kumari et al., 2025). Similarly, Li et al. (2023a) developed an IoT system in the Raja Musa Forest Reserve, Malaysia, and proposed a low-complexity model to calculate the Duff Moisture Code using groundwater level and relative humidity, achieving a correlation of 0.91 with actual values. Li et al. (2023b) further demonstrated that multi-layer neural networks can predict FWI from IoT-measured fire risk factors, where a nine-input five-layer network outperformed simpler models with an mean square error of 1.116 and a correlation of 0.890. In addition, Li et al. (2022) showed that groundwater level can be incorporated into tropical peatland FWI formulation by replacing the Drought Code inputs of temperature and rainfall with groundwater level and validating the approach using LoRa-based IoT data and machine learning prediction. These IoT- and index-driven approaches are useful for identifying local drying conditions and supporting site specific early warning. However, their application over extensive tropical peatland landscapes may be constrained by sensor deployment density, maintenance requirements, network connectivity, and regional calibration. The proposed framework complements these approaches by using spatially continuous gridded meteorological, hydrological, land cover, anthropogenic, and historical fire data to learn large area spatio-temporal fire risk patterns without requiring dense in-situ sensor networks.

603 This advantage is reflected in the resulting probability of fire maps (**Figure 9**), which
604 represent fire likelihood across the study area. The probability based output provides a more
605 informative representation compared to categorical fire danger levels (Mahdiyasa et al., 2025;
606 Mashuri et al., 2026; Taufik et al., 2022), because it indicates not only areas with elevated fire
607 risk but also the relative likelihood of occurrence across the landscape. This allows the
608 proposed model to distinguish areas with different levels of susceptibility and to represent
609 gradual variation in fire risk. Furthermore, the agreement between predicted fire prone areas
610 and the observed hotspot distribution (**Figure 11**) suggests that the proposed model is able to
611 capture the dominant patterns of tropical peatland fire occurrence and translate them into
612 reliable probability estimates.
613
614 **4.1 Forecasting and early detection of tropical peatland fire**
615 The forecasting results (**Figure 11**) indicate that the location and extent of susceptible
616 areas vary across prediction horizons, suggesting that the model captures the evolution of fire
617 prone conditions under changing environmental conditions. This behaviour is important in
618 tropical peatlands, where fire susceptibility is influenced by short-term drying, prolonged
619 moisture deficits, and seasonal variability (Harrison et al., 2024; Mezbahuddin et al., 2023;
620 Ramdzan et al., 2023). Under these conditions, areas that initially exhibit low probability of
621 fire may become increasingly vulnerable over time, while other areas may indicate declining
622 susceptibility. Consequently, the model represents fire risk as a dynamic process that changes
623 with hydrological and meteorological conditions. This capability increases the practical
624 relevance of the model for early warning by capturing both current fire risk and its probability
625 evolution over time.
626 From an operational perspective, the forecasting capability of the proposed model is
627 valuable for early detection and targeted fire management in tropical peatland regions (Horton

Journal Pre-proof

et al., 2022; S. Mishra et al., 2021). By anticipating the expansion or contraction of susceptible zones over time, the model provides actionable information for monitoring and mitigation processes. Areas projected to experience decreasing susceptibility may be assigned lower priority for operational response. In contrast, areas with persistently high fire probability may be monitored as critical zones requiring continuous attention. As a result, the model not only produces fire risk maps but also supports a more adaptive approach to peatland fire management.

4.2 Sensitivity of tropical peatland fire probability maps to different climate conditions

Wetter and cooler climate conditions reduce tropical peatland fire risk by enhancing moisture retention within the peat pore space (Horton et al., 2022; Taufik et al., 2022). Higher rainfall, increased soil moisture, and lower air temperature contribute to wetter peat conditions, reducing the likelihood of ignition and limiting fire spread across peatland areas. Under these conditions, peat is less vulnerable to sustained burning, and the extent of areas with elevated fire susceptibility could decrease. These effects can reduce both the probability of fire occurrence and the spatial persistence of high-risk zones across the landscape. A corresponding spatial pattern is the expansion of areas classified as having zero probability of fire occurrence, with proportions increasing from about 22% to 42% for the next three-day forecast (**Figure 13a, b, c)** and from about 18% to 35% for the next ten-day forecast (**Figure 14a, b, c)** across perturbation levels from 10% to 50%. Consequently, a larger portion of the peatland landscape becomes effectively resilient to fire under wetter and cooler conditions.

In contrast, drier and warmer climate conditions increase tropical peatland fire risk by intensifying peat drying and reducing moisture availability (Kuwata, 2024; Mezbahuddin et al., 2023; Schmidt et al., 2024). A combination of decreased rainfall, lower soil moisture, and higher air temperature accelerate the loss of water from the peat surface and enlarge the

unsaturated zone, creating conditions that are more favourable for ignition and fire spread. Under these conditions, peat becomes more vulnerable to persistent burning, particularly during periods of prolonged moisture deficit. In addition to increasing the probability of fire occurrence, these effects also enlarge the extent and persistence of susceptible areas across the landscape. A notable pattern is the substantial reduction in areas classified as having zero probability of fire occurrence, with proportions decreasing from about 10% to 1% for the next three-day forecast (**Figure 13d, e, f)** and from about 8% to 0% for the next ten-day forecast (**Figure 14d, e, f)** over perturbation levels ranging from 10% to 50%. This pattern indicates that regions previously considered effectively free from fire risk become increasingly susceptible under drier and warmer conditions. As a result, drier conditions could lead to more widespread and severe fire prone conditions in tropical peatlands.

4.3 Limitations and future research

The application of isotonic regression for calibration purposes depends on the availability and representativeness of the calibration data. In highly imbalanced tropical peatland fire datasets, where fire events are relatively rare, isotonic regression may be sensitive to sparse observations (Tian & Zhang, 2022; Zhang et al., 2025). This condition could produce stepwise calibration functions that are unstable and may not be consistent under different spatial or temporal conditions. Future research could develop calibration models that vary across regions, seasons, or peatland conditions (Little et al., 2024; Tohme et al., 2020). This would allow the calibrated probabilities to more accurately represent fire risk characteristics and changing environmental conditions, which improves the reliability of probability of fire maps for operational early warning systems.

Although the proposed model can learn spatial and temporal patterns from meteorological, hydrological, land cover, anthropogenic, and fire data, it may not fully capture

Journal Pre-proof

peatland structural changes, including variations in bulk density, active porosity, and hydraulic conductivity. These properties are important because they affect water retention, groundwater dynamics, peat drying, and the susceptibility of peat to fire (Baird et al., 2017; Crawford et al., 2024; Rezanezhad et al., 2016; Wijedasa, 2016). Future research could integrate the proposed data-driven framework with physics-based peatland models to provide a more process-based representation of tropical peatland fire risk (Kurnianto et al., 2015; Mahdiyasa et al., 2022, 2023, 2024; Young et al., 2023). Furthermore, the framework should also incorporate land use dynamics, critical variables in peatland conversion that are closely linked to increased peat fire risks. This integration would allow the model to account not only for observed fire patterns but also for the ecohydrological and mechanical processes that regulate peatland condition. By incorporating changes in peat physical properties and hydrological behavior, future models could improve the interpretation of fire risk dynamics and support more reliable long-term forecasting under different environmental and land management scenarios.

The spatial resolution used in this study may influence the representation of small fire patches. All input datasets were resampled to a uniform 5 × 5 km grid to ensure spatial consistency among data sources with different native resolutions. Similar grid sizes have been used in previous fire related studies (Bar et al., 2023; Tan et al., 2020; Wickramasinghe et al., 2024), indicating that the selected resolution is suitable for regional scale fire risk mapping and for integrating meteorological, hydrological, land cover, anthropogenic, and fire occurrence variables within a common modeling framework. However, the 5 km grid should be interpreted as a grid cell level fire occurrence probability unit rather than a fine scale fire patch detection resolution. In this study, a grid cell was labeled as positive when it contained at least one hotspot detection during the corresponding time step. Consequently, the model estimates the probability of fire occurrence within each grid cell rather than delineating the exact size, shape, or boundary of individual fire patches. Small or isolated fire events may be aggregated with

surrounding non-fire conditions, which can smooth fine scale spatial variability and reduce the ability of the model to represent localized fire clusters. Further research should examine the sensitivity of model performance to alternative grid resolutions and evaluate their effects on hotspot.

Future could also develop the proposed model into an operational dashboard for a tropical peatland fire early warning system. This development would allow users to monitor fire risk conditions dynamically, identify areas with increasing fire probability, and support timely prevention or mitigation actions (Horton et al., 2022). The dashboard might include alert levels based on calibrated fire probabilities, enabling the model outputs to be translated into practical decision support information for land managers, government agencies, and plantation operators (Purnomo et al., 2024). Therefore, the development should focus on real time data integration and user-friendly visualization to ensure that the dashboard is reliable for operational tropical peatland fire management.

5. Conclusion

This study developed a spatio-temporal framework to analyze and predict tropical peatland fire risk by integrating a Convolutional Neural Network (CNN) for spatial feature extraction, Convolutional Long Short-Term Memory (ConvLSTM) for temporal dependency learning, and isotonic regression for probability calibration. Through this integrated structure, the proposed model is able to identify areas with increased fire probability and produces probability of fire maps that are consistent with observed hotspot distributions. Furthermore, the probability calibration method employed by this framework improve the agreement between model outputs and observed fire event frequencies, which support risk assessment in tropical peatland ecosystems. Therefore, the proposed model provides not only a predictive

Journal Pre-proof

tool but also decision support information for improving tropical peatland fire mitigation and restoration management.

CRediT authorship contribution statement

Adilan Mahdiyasa: Conceptualization, Methodology, Software, Formal analysis, Writing-original draft, Writing-review & editing.

Jeremy Chaim: Software, Methodology, Formal analysis, Writing-review & editing.

Udjianna Pasaribu: Methodology, Writing-review & editing.

Elham Sumarga: Methodology, Writing-review & editing.

Bagus Muljadi: Methodology, Resources, Writing-review & editing.

Asrini Chrysanti: Methodology, Writing-review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Peatland distribution data are obtained from Geospatial Information Agency of Indonesia (BIG) at https://geoportal.big.go.id/#/. Fire hotspots data are obtained from Fire Information for Resource Management System (FIRMS) at https://firms.modaps.eosdis.nasa.gov/. Rainfall data are obtained from Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) at https://www.chc.ucsb.edu/data/chirps. Air temperature and soil moisture data are obtained from ERA5-Land reanalysis dataset at https://cds.climate.copernicus.eu/. Land cover

Journal Pre-proof

classification data are obtained from MODIS Collection 6 Land Cover Type Product at https://modis.gsfc.nasa.gov/data/dataprod/mod12.php. Distance to human activity data are obtained from Global Human Settlement Layer (GHSL) at https://human-settlement.emergency.copernicus.eu/.

Acknowledgments

This research is funded by the ITB Research Program 2026 under ITB International Research Scheme through the Directorate of Research and Innovation, Institut Teknologi Bandung (Project ID: FMIPA.PN-6-123-2026).

Journal Pre-proof

Journal Pre-proof

References

Alexandrov, G. A., Brovkin, V. A., Kleinen, T., & Yu, Z. (2020). The capacity of northern peatlands for long-term carbon sequestration. *Biogeosciences*, *17*(1), 47–54. https://doi.org/10.5194/bg-17-47-2020

Anda, M., Ritung, S., Suryani, E., Sukarman, Hikmat, M., Yatno, E., Mulyani, A., Subandiono, R. E., Suratman, & Husnain. (2021). Revisiting tropical peatlands in Indonesia: Semi-detailed mapping, extent and depth distribution assessment. *Geoderma*, *402*, 115235. https://doi.org/10.1016/j.geoderma.2021.115235

Baird, A. J., Low, R., Young, D., Swindles, G. T., Lopez, O. R., & Page, S. (2017). High permeability explains the vulnerability of the carbon store in drained tropical peatlands. *Geophysical Research Letters*, *44*(3), 1333–1339. https://doi.org/10.1002/2016GL072245

Bar, S., Parida, B. R., Pandey, A. C., Shankar, B. U., Kumar, P., Panda, S. K., & Behera, M. D. (2023). Modeling and prediction of fire occurrences along an elevational gradient in Western Himalayas. *Applied Geography*, *151*, 102867. https://doi.org/10.1016/j.apgeog.2022.102867

Crawford, A. J., Belcher, C. M., New, S., Gallego-Sala, A., Swindles, G. T., Page, S., Blyakharchuk, T. A., Cadillo-Quiroz, H., Charman, D. J., Gałka, M., Hughes, P. D. M., Lähteenoja, O., Mauquoy, D., Roland, T. P., & Väliranta, M. (2024). Tropical peat composition may provide a negative feedback on fire occurrence and severity. *Nature Communications*, *15*(1), 7363. https://doi.org/10.1038/s41467-024-50916-7

Delong, Ł., & V. Wüthrich, M. (2025). Isotonic Regression for Variance Estimation and Its Role in Mean Estimation and Model Validation. *North American Actuarial Journal*, *29*(3), 563–591. https://doi.org/10.1080/10920277.2024.2421221

Journal Pre-proof

784 Farghaly, A., Owais, M., Sattar, A. M. A., & Elhadidy, M. (2026). Interpretable machine
785 learning and rolling-origin validation identify two distinct control and predictability
786 regimes in full-scale aerated lagoon performance. *Journal of Environmental*
787 *Management*, *413*, 130281. https://doi.org/10.1016/j.jenvman.2026.130281
788 Fauziah, Prasetyo, L. B., Saribanon, N., & Hayati, N. (2025). Vulnerability of peatland fires in
789 bengkalis regency during the ENSO El nino phase using a machine learning approach.
790 *MethodsX*, *14*, 103128. https://doi.org/10.1016/j.mex.2024.103128
791 Felton, A. M., Engström, L. M., Felton, A., & Knott, C. D. (2003). Orangutan population
792 density, forest structure and fruit availability in hand-logged and unlogged peat
793 swamp forests in West Kalimantan, Indonesia. *Biological Conservation*, *114*(1), 91–
794 101. https://doi.org/10.1016/S0006-3207(03)00013-2
795 Fiorucci, J. A., & Louzada, F. (2020). GROEC: Combination method via Generalized Rolling
796 Origin Evaluation. *M4 Competition*, *36*(1), 105–109.
797 https://doi.org/10.1016/j.ijforecast.2019.04.013
798 Funk, C., Peterson, P., Landsfeld, M., Pedreros, D., Verdin, J., Shukla, S., Husak, G., Rowland,
799 J., Harrison, L., Hoell, A., & Michaelsen, J. (2015). The climate hazards infrared
800 precipitation with stations—A new environmental record for monitoring extremes.
801 *Scientific Data*, *2*(1), 150066. https://doi.org/10.1038/sdata.2015.66
802 Geospatial Information Agency. (2019). *Peta Rupabumi Digital Indonesia* [Dataset].
803 https://geoportal.big.go.id/#/
804 Giglio, L., Schroeder, W., & Justice, C. O. (2016). The collection 6 MODIS active fire detection
805 algorithm and fire products. *Remote Sensing of Environment*, *178*, 31–41.
806 https://doi.org/10.1016/j.rse.2016.02.054

807 Girkin, N. T., Cooper, H. V., Ledger, M. J., O'Reilly, P., Thornton, S. A., Åkesson, C. M., Cole, L.
808 E. S., Hapsari, K. A., Hawthorne, D., & Roucoux, K. H. (2022). Tropical peatlands in the
809 Anthropocene: The present and the future. *Anthropocene*, *40*, 100354.
810 https://doi.org/10.1016/j.ancene.2022.100354
811 Gneiting, T., & Resin, J. (2023). Regression diagnostics meets forecast evaluation: Conditional
812 calibration, reliability diagrams, and coefficient of determination. *Electronic Journal*
813 *of Statistics*, *17*(2), 3226–3286. https://doi.org/10.1214/23-EJS2180
814 Guria, R., Mishra, M., Mohanta, S., & Paul, S. (2025). Forest fire probability zonation using
815 dNBR and machine learning models: A case study at the Similipal Biosphere
816 Reserve (SBR), Odisha, India. *Environmental Science and Pollution Research*, *32*(59),
817 31375–31396. https://doi.org/10.1007/s11356-025-35976-6
818 Guria, R., Mishra, M., Silva, R. M. da, Mishra, M., & Santos, C. A. G. (2024). Predicting forest
819 fire probability in Similipal Biosphere Reserve (India) using Sentinel-2 MSI data and
820 machine learning. *Remote Sensing Applications: Society and Environment*, *36*,
821 101311. https://doi.org/10.1016/j.rsase.2024.101311
822 Hapfelmeier, A., Hornung, R., & Haller, B. (2023). Efficient permutation testing of variable
823 importance measures by the example of random forests. *Computational Statistics &*
824 *Data Analysis*, *181*, 107689. https://doi.org/10.1016/j.csda.2022.107689
825 Harrison, M. E., Deere, N. J., Imron, M. A., Nasir, D., Asti, H. A., Aragay Soler, J., Boyd, N. C.,
826 Cheyne, S. M., Collins, S. A., D'Arcy, L. J., Erb, W. M., Green, H., Healy, W., Holly, B.,
827 Houlihan, P. R., Husson, S. J., Jeffers, K. A., Kulu, I. P., Kusin, K., … Struebig, M. J.
828 (2024). Impacts of fire and prospects for recovery in a tropical peat forest ecosystem.
829 *Proceedings of the National Academy of Sciences*, *121*(17), e2307216121.
830 https://doi.org/10.1073/pnas.2307216121

Journal Pre-proof

831 Hein, L., Spadaro, J. V., Ostro, B., Hammer, M., Sumarga, E., Salmayenti, R., Boer, R., Tata, H.,
832 Atmoko, D., & Castañeda, J.-P. (2022). The health impacts of Indonesian peatland
833 fires. *Environmental Health*, *21*(1), 62. https://doi.org/10.1186/s12940-022-00872-w
834 Horton, A. J., Lehtinen, J., & Kummu, M. (2022). Targeted land management strategies could
835 halve peatland fire occurrences in Central Kalimantan, Indonesia. *Communications*
836 *Earth & Environment*, *3*(1), 204. https://doi.org/10.1038/s43247-022-00534-2
837 Horton, A. J., Virkki, V., Lounela, A., Miettinen, J., Alibakhshi, S., & Kummu, M. (2021).
838 Identifying Key Drivers of Peatland Fires Across Kalimantan's Ex-Mega Rice Project
839 Using Machine Learning. *Earth and Space Science*, *8*(12), e2021EA001873.
840 https://doi.org/10.1029/2021EA001873
841 Khalaf, Mhd. W. A., Jouibary, S. S., & Jahdi, R. (2024). Performance Analysis of ConvLSTM,
842 FlamMap, and CA Algorithms to Predict Wildfire Spread in Golestan National Park, NE
843 Iran. *Environmental Modeling & Assessment*, *29*(3), 489–502.
844 https://doi.org/10.1007/s10666-024-09956-y
845 Khirani, S., Souahlia, A., Rabehi, A., Bourennane, M., Guermoui, M., Tibermacine, I. E., &
846 Rabehi, A. (2026). Advanced evaluation of pre-trained CNN models for accurate
847 forest fire detection. *Natural Hazards*, *122*(6), 234. https://doi.org/10.1007/s11069-
848 026-07976-3
849 Kiely, L., Spracklen, D. V., Arnold, S. R., Papargyropoulou, E., Conibear, L., Wiedinmyer, C.,
850 Knote, C., & Adrianto, H. A. (2021). Assessing costs of Indonesian fires and the
851 benefits of restoring peatland. *Nature Communications*, *12*(1), 7044.
852 https://doi.org/10.1038/s41467-021-27353-x
853 Koplitz, S. N., Mickley, L. J., Marlier, M. E., Buonocore, J. J., Kim, P. S., Liu, T., Sulprizio, M. P.,
854 DeFries, R. S., Jacob, D. J., Schwartz, J., Pongsiri, M., & Myers, S. S. (2016). Public

Journal Pre-proof

855 health impacts of the severe haze in Equatorial Asia in September–October 2015:
856 Demonstration of a new framework for informing fire management strategies to
857 reduce downwind smoke exposure. *Environmental Research Letters*, *11*(9), 094023.
858 https://doi.org/10.1088/1748-9326/11/9/094023
859 Kumari, P. P., Vidhura, P., Eshwari, A., Shankar, K. J., Mohan, G. V. K., & Reddy, V. V. K. (2025).
860 IOT - Enabled Machine learning for Ground Water Level monitoring in peatlands.
861 *International Journal of Computational Learning & Intelligence*, *4*(4).
862 https://doi.org/10.5281/zenodo.15251503
863 Kurnianto, S., Warren, M., Talbot, J., Kauffman, B., Murdiyarso, D., & Frolking, S. (2015).
864 Carbon accumulation of tropical peatlands over millennia: A modeling approach.
865 *Global Change Biology*, *21*(1), 431–444. https://doi.org/10.1111/gcb.12672
866 Kurniawan, R., Muttaqien, F. H., Syafarina, I., Belgaman, H. A., Dewi, S., & Latifah, A. L.
867 (2025). Data-driven model for predicting peatland fire hotspots and carbon emissions
868 in South Kalimantan. *Wetlands Ecology and Management*, *33*(3), 44.
869 https://doi.org/10.1007/s11273-025-10058-z
870 Kuwata, M. (2024). Atmospheric emissions, processes, and impacts of tropical peatland fire
871 haze in Equatorial Asia: A review. *Atmospheric Environment*, *331*, 120575.
872 https://doi.org/10.1016/j.atmosenv.2024.120575
873 Ledger, M. J., Sowter, A., Morrison, K., Evans, C. D., Large, D. J., Athab, A., Gee, D., Brown, C.,
874 & Sjögersten, S. (2024). Potential of APSIS-InSAR for measuring surface oscillations of
875 tropical peatlands. *PLOS ONE*, *19*(2), e0298939.
876 https://doi.org/10.1371/journal.pone.0298939

Journal Pre-proof

Li, L. (2025). A Comprehensive Survey of Fire Weather Index (FWI) Systems and IoT Applications in Peatland Fire Management. *IEEE Access*, *13*, 33579–33599. https://doi.org/10.1109/ACCESS.2025.3543570

Li, L., A. Sali, J. T. Liew, N. L. Saleh, S. M. S. Ahmad, A. M. Ali, A. A. Nuruddin, N. A. Aziz, I. S. Sitanggang, L. Syaufina, A. D. Nurhayati, H. Nishino, & N. Asai. (2022). Estimation of Ground Water Level (GWL) for Tropical Peatland Forest Using Machine Learning. *IEEE Access*, *10*, 126180–126187. https://doi.org/10.1109/ACCESS.2022.3225906

Li, L., Sali, A., Noordin, N. K., Ismail, A., & Hashim, F. (2023b). Prediction of Peatlands Forest Fires in Malaysia Using Machine Learning. *Forests*, *14*(7), 1472. https://doi.org/10.3390/f14071472

Li, L., Sali, A., Noordin, N. K., Ismail, A., Hashim, F., Rasid, M. F. A., Hanafi, M., Razali, S. M., Aziz, N. A., Sukaesih Sitanggang, I., Syaufina, L., & Nurhayati, A. D. (2023a). Modeling of Evaporation Rate for Peatland Fire Prevention Using Internet of Things (IoT) System. *Fire*, *6*(7), 272. https://doi.org/10.3390/fire6070272

Little, M. P., Hamada, N., & Zablotska, L. B. (2024). A generalisation of the method of regression calibration and comparison with Bayesian and frequentist model averaging methods. *Scientific Reports*, *14*(1), 6613. https://doi.org/10.1038/s41598-024-56967-6

Loisel, J., Gallego-Sala, A. V., Amesbury, M. J., Magnan, G., Anshari, G., Beilman, D. W., Benavides, J. C., Blewett, J., Camill, P., Charman, D. J., Chawchai, S., Hedgpeth, A., Kleinen, T., Korhola, A., Large, D., Mansilla, C. A., Müller, J., van Bellen, S., West, J. B., … Wu, J. (2021). Expert assessment of future vulnerability of the global peatland carbon sink. *Nature Climate Change*, *11*(1), 70–77. https://doi.org/10.1038/s41558-020-00944-0

Journal Pre-proof

Lupascu, M., Varkkey, H., & Tortajada, C. (2020). Is flooding considered a threat in the degraded tropical peatlands? *Science of The Total Environment*, *723*, 137988. https://doi.org/10.1016/j.scitotenv.2020.137988

Mahdiyasa, A. W., Large, D. J., Icardi, M., & Muljadi, B. P. (2024). MPeat2D – a fully coupled mechanical–ecohydrological model of peatland development in two dimensions. *Earth Surface Dynamics*, *12*(4), 929–952. https://doi.org/10.5194/esurf-12-929-2024

Mahdiyasa, A. W., Large, D. J., Muljadi, B. P., & Icardi, M. (2023). Modelling the influence of mechanical-ecohydrological feedback on the nonlinear dynamics of peatlands. *Ecological Modelling*, *478*, 110299. https://doi.org/10.1016/j.ecolmodel.2023.110299

Mahdiyasa, A. W., Large, D. J., Muljadi, B. P., Icardi, M., & Triantafyllou, S. (2022). MPeat—A fully coupled mechanical-ecohydrological model of peatland development. *Ecohydrology*, *15*(1), e2361. https://doi.org/10.1002/eco.2361

Mahdiyasa, A. W., Melly, M., Pasaribu, U. S., Taufik, M., & Muljadi, B. P. (2025). Peatfr: An R package to forecast tropical peatland fire risk with stochastic, machine learning, and optimisation methods. *Ecological Informatics*, *92*, 103532. https://doi.org/10.1016/j.ecoinf.2025.103532

Mambile, C., Kaijage, S., & Leo, J. (2025). Deep learning models for enhanced forest-fire prediction at Mount Kilimanjaro, Tanzania: Integrating satellite images, weather data and human activities data. *Natural Hazards Research*, *5*(2), 335–347. https://doi.org/10.1016/j.nhres.2024.12.001

Mashuri, Karlina, & Sujono, J. (2026). Development of Keetch and Byram Drought Index for future projection in the Siak Watershed, Indonesia. *Natural Hazards Research*. https://doi.org/10.1016/j.nhres.2026.05.005

Journal Pre-proof

Mehdiyev, N., Majlatow, M., & Fettke, P. (2025). Integrating permutation feature importance with conformal prediction for robust Explainable Artificial Intelligence in predictive process monitoring. *Engineering Applications of Artificial Intelligence*, *149*, 110363. https://doi.org/10.1016/j.engappai.2025.110363

Mezbahuddin, S., Nikonovas, T., Spessa, A., Grant, R. F., Imron, M. A., Doerr, S. H., & Clay, G. D. (2023). Accuracy of tropical peat and non-peat fire forecasts enhanced by simulating hydrology. *Scientific Reports*, *13*(1), 619. https://doi.org/10.1038/s41598-022-27075-0

Mishra, M., Guria, R., Baraj, B., Nanda, A. P., Santos, C. A. G., Silva, R. M. da, & Laksono, F. A. T. (2024). Spatial analysis and machine learning prediction of forest fire susceptibility: A comprehensive approach for effective management and mitigation. *Science of The Total Environment*, *926*, 171713. https://doi.org/10.1016/j.scitotenv.2024.171713

Mishra, M., Guria, R., Baraj, B., Santos, C. A. G., Silva, R. M. da, dos Santos, C. A. C., & Bhutia, K. D. O. (2025). Climate variability and forest fires: Trends, correlation, spatiotemporal patterns in the Seven Sister States of northeastern India (2001–2022). *Remote Sensing Applications: Society and Environment*, *38*, 101586. https://doi.org/10.1016/j.rsase.2025.101586

Mishra, S., Page, S. E., Cobb, A. R., Lee, J. S. H., Jovani-Sancho, A. J., Sjögersten, S., Jaya, A., Aswandi, & Wardle, D. A. (2021). Degradation of Southeast Asian tropical peatlands and integrated strategies for their better management and restoration. *Journal of Applied Ecology*, *58*(7), 1370–1387. https://doi.org/10.1111/1365-2664.13905

Mukhaiyar, U., Mahdiyasa, A. W., Prastoro, T., Pasaribu, U. S., Sari, K. N., Indratno, S. W., Soekarno, I., Choesin, D. N., Ismail, I., Rosleine, D., & Qoyyimi, D. T. (2024). The generalized STAR modelling with three-dimensional of spatial weight matrix in

Journal Pre-proof

predicting the Indonesia peatland's water level. *Environmental Sciences Europe*, *36*(1), 180. https://doi.org/10.1186/s12302-024-00979-6

Muñoz-Sabater, J., Dutra, E., Agustí-Panareda, A., Albergel, C., Arduini, G., Balsamo, G., Boussetta, S., Choulga, M., Harrigan, S., Hersbach, H., Martens, B., Miralles, D. G., Piles, M., Rodríguez-Fernández, N. J., Zsoter, E., Buontempo, C., & Thépaut, J.-N. (2021). ERA5-Land: A state-of-the-art global reanalysis dataset for land applications. *Earth Syst. Sci. Data*, *13*(9), 4349–4383. https://doi.org/10.5194/essd-13-4349-2021

Nie, L., Wang, L., Li, Y., Yang, P., Lv, Z., Liu, Y., Xu, W., Ou, G., & Kou, W. (2026). Forest fire risk zonation in Yunnan Province using multi-source indicators and interpretable machine learning. *Ecological Informatics*, *95*, 103769. https://doi.org/10.1016/j.ecoinf.2026.103769

Page, S., Mishra, S., Agus, F., Anshari, G., Dargie, G., Evers, S., Jauhiainen, J., Jaya, A., Jovani-Sancho, A. J., Laurén, A., Sjögersten, S., Suspense, I. A., Wijedasa, L. S., & Evans, C. D. (2022). Anthropogenic impacts on lowland tropical peatland biogeochemistry. *Nature Reviews Earth & Environment*, *3*(7), 426–443. https://doi.org/10.1038/s43017-022-00289-6

Pesaresi, M., Schiavina, M., Politis, P., Freire, S., Krasnodębska, K., Uhl, J. H., Carioli, A., Corbane, C., Dijkstra, L., Florio, P., Friedrich, H. K., Gao, J., Leyk, S., Lu, L., Maffenini, L., Mari-Rivero, I., Melchiorri, M., Syrris, V., Van Den Hoek, J., & Kemper, T. (2024). Advances on the Global Human Settlement Layer by joint assessment of Earth Observation and population survey data. *International Journal of Digital Earth*, *17*(1), 2390454. https://doi.org/10.1080/17538947.2024.2390454

Purnomo, H., Puspitaloka, D., Okarda, B., Andrianto, A., Qomar, N., Sutikno, S., Muhammad, A., Basuki, I., Jalil, A., Yesi, Prasetyo, P., Tarsono, Zulkardi, Kusumadewi, S. D.,

973 Komarudin, H., Dermawan, A., & Brady, M. A. (2024). Community-based fire
974 prevention and peatland restoration in Indonesia: A participatory action research
975 approach. *Environmental Development*, *50*, 100971.
976 https://doi.org/10.1016/j.envdev.2024.100971
977 Ramdzan, K. N. M., Moss, P. T., Jacobsen, G., Gallego-Sala, A., Charman, D., Harrison, M. E.,
978 Page, S., Mishra, S., Wardle, D. A., Jaya, A., Aswandi, Nasir, D., & Yulianti, N. (2023).
979 Insights for restoration: Reconstructing the drivers of long-term local fire events and
980 vegetation turnover of a tropical peatland in Central Kalimantan. *Palaeogeography,*
981 *Palaeoclimatology, Palaeoecology*, *628*, 111772.
982 https://doi.org/10.1016/j.palaeo.2023.111772
983 Rezanezhad, F., Price, J. S., Quinton, W. L., Lennartz, B., Milojevic, T., & Van Cappellen, P.
984 (2016). Structure of peat soils and implications for water storage, flow and solute
985 transport: A review update for geochemists. *Chemical Geology*, *429*, 75–84.
986 https://doi.org/10.1016/j.chemgeo.2016.03.010
987 Rodríguez Vásquez, M. J., Benoist, A., Roda, J.-M., & Fortin, M. (2021). Estimating
988 Greenhouse Gas Emissions From Peat Combustion in Wildfires on Indonesian
989 Peatlands, and Their Uncertainty. *Global Biogeochemical Cycles*, *35*(2),
990 e2019GB006218. https://doi.org/10.1029/2019GB006218
991 Schmidt, A., Ellsworth, L. M., Boisen, G. A., Novita, N., Malik, A., Gangga, A., Albar, I., Dwi
992 Nurhayati, A., Putra Ritonga, R., Asyhari, A., & Kauffman, J. B. (2024). Fire frequency,
993 intensity, and burn severity in Kalimantan's threatened Peatland areas over two
994 Decades. *Frontiers in Forests and Global Change*, *Volume 7-2024*.
995 https://www.frontiersin.org/journals/forests-and-global-
996 change/articles/10.3389/ffgc.2024.1221797

Journal Pre-proof

997 Situngkir, A. O., Mahdiyasa, A. W., Sari, K. N., & Pasaribu, U. S. (2025). Modelling Peatland
998 Fire Risk and Economic Losses in Indonesia. *2025 5th International Conference on*
999 *Innovative Research in Applied Science, Engineering and Technology (IRASET)*, 1–6.
1000 https://doi.org/10.1109/IRASET64571.2025.11008022
1001 Sulla-Menashe, D., Gray, J. M., Abercrombie, S. P., & Friedl, M. A. (2019). Hierarchical
1002 mapping of annual global land cover 2001 to present: The MODIS Collection 6 Land
1003 Cover product. *Remote Sensing of Environment*, *222*, 183–194.
1004 https://doi.org/10.1016/j.rse.2018.12.013
1005 Tan, Z. D., Carrasco, L. R., & Taylor, D. (2020). Spatial correlates of forest and land fires in
1006 Indonesia. *International Journal of Wildland Fire*, *29*(12), 1088–1099.
1007 https://doi.org/10.1071/WF20036
1008 Taufik, M., Widyastuti, M. T., Sulaiman, A., Murdiyarso, D., Santikayasa, I. P., & Minasny, B.
1009 (2022). An improved drought-fire assessment for managing fire risks in tropical
1010 peatlands. *Agricultural and Forest Meteorology*, *312*, 108738.
1011 https://doi.org/10.1016/j.agrformet.2021.108738
1012 Taufik, Muh., Setiawan, B. I., & van Lanen, H. A. J. (2015). Modification of a fire drought
1013 index for tropical wetland ecosystems by including water table depth. *Agricultural*
1014 *and Forest Meteorology*, *203*, 1–10.
1015 https://doi.org/10.1016/j.agrformet.2014.12.006
1016 Tian, Y., & Zhang, Y. (2022). A comprehensive survey on regularization strategies in machine
1017 learning. *Information Fusion*, *80*, 146–166.
1018 https://doi.org/10.1016/j.inffus.2021.11.005

Journal Pre-proof

1019 Tohme, T., Vanslette, K., & Youcef-Toumi, K. (2020). A generalized Bayesian approach to
1020 model calibration. *Reliability Engineering & System Safety*, *204*, 107141.
1021 https://doi.org/10.1016/j.ress.2020.107141
1022 Wang, Z., Fang, K., & Yao, Q. (2022). Fire history and its forcing in Northeastern Asia boreal
1023 forests. *Natural Hazards Research*, *2*(3), 166–171.
1024 https://doi.org/10.1016/j.nhres.2022.07.001
1025 Wickramasinghe, A. M. K., Boer, M. M., Cunningham, C. X., Nolan, R. H., Bowman, D. M. J. S.,
1026 & Williamson, G. J. (2024). Modeling the Probability of Dry Lightning-Induced
1027 Wildfires in Tasmania: A Machine Learning Approach. *Geophysical Research Letters*,
1028 *51*(16), e2024GL110381. https://doi.org/10.1029/2024GL110381
1029 Wijedasa, L. S. (2016). Peat soil bulk density important for estimation of peatland fire
1030 emissions. *Global Change Biology*, *22*(9), 2959–2959.
1031 https://doi.org/10.1111/gcb.13364
1032 Wu, H., Gao, H., Huang, Y., & Xu, C. (2026). Analysis of the Compound Disaster Caused by
1033 Extreme Rainfall and Landslides in Sumatra, Indonesia in 2025 and Its Implications for
1034 Disaster Prevention and Mitigation. *Natural Hazards Research*.
1035 https://doi.org/10.1016/j.nhres.2026.03.003
1036 Xu, Z., Li, J., Cheng, S., Rui, X., Zhao, Y., He, H., Guan, H., Sharma, A., Erxleben, M., Chang, R.,
1037 & Xu, L. L. (2025). Deep learning for wildfire risk prediction: Integrating remote
1038 sensing and environmental data. *ISPRS Journal of Photogrammetry and Remote*
1039 *Sensing*, *227*, 632–677. https://doi.org/10.1016/j.isprsjprs.2025.06.002
1040 Y. Ho & S. Wookey. (2020). The Real-World-Weight Cross-Entropy Loss Function: Modeling
1041 the Costs of Mislabeling. *IEEE Access*, *8*, 4806–4813.
1042 https://doi.org/10.1109/ACCESS.2019.2962617

Journal Pre-proof

Yar, H., Ullah, W., Ahmad Khan, Z., & Wook Baik, S. (2023). An Effective Attention-based CNN Model for Fire Detection in Adverse Weather Conditions. *ISPRS Journal of Photogrammetry and Remote Sensing*, *206*, 335–346. https://doi.org/10.1016/j.isprsjprs.2023.10.019

Young, D. M., Baird, A. J., Morris, P. J., Dargie, G. C., Mampouya Wenina, Y. E., Mbemba, M., Boom, A., Cook, P., Betts, R., Burke, E., Bocko, Y. E., Chadburn, S., Crabtree, D. E., Crezee, B., Ewango, C. E. N., Garcin, Y., Georgiou, S., Girkin, N. T., Gulliver, P., … Lewis, S. L. (2023). Simulating carbon accumulation and loss in the central Congo peatlands. *Global Change Biology*, *29*(23), 6812–6827. https://doi.org/10.1111/gcb.16966

Zelioli, L., Farahnakian, F., Middleton, M., Pitkänen, T. P., Tuominen, S., Nevalainen, P., Pohjankukka, J., & Heikkonen, J. (2025). Peatland pixel-level classification via multispectral, multiresolution and multisensor data using convolutional neural network. *Ecological Informatics*, *90*, 103233. https://doi.org/10.1016/j.ecoinf.2025.103233

Zhang, J., Wang, Y., Wang, M., & Wang, B. (2025). Sparse and smooth additive isotonic model in high-dimensional settings. *Machine Learning*, *114*(3), 50. https://doi.org/10.1007/s10994-024-06641-9

Journal Pre-proof

Journal Pre-proof

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The author *Click here to enter your name* is Choose an item for *Click here to enter the journal's name* and was not involved in the editorial review or the decision to publish this article.

☐The authors declare the following financial interests (e.g., any funding for the research project)/personal relationships (e.g., the author is an employee of a profitable company) which may be considered as potential competing interests:

Click here to enter your full declaration

Journal Pre-proof